When Inflation Turns Volatile: Expectations of Households and Professionals[†]

Yi-Hsuan Chang Chien Nan-Kuang Chen Tsung-Hsien Li[*]

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Abstract

Are professional inflation forecasts more accurate than household expectations, and does the way each group forms expectations change with the inflation environment? We use median one-year-ahead inflation expectations from the University of Michigan Survey of Consumers, the Survey of Professional Forecasters, and the Livingston Survey over 1981–2024. A two-state Markov-switching model of realized consumer price inflation divides the sample into stable and volatile regimes. We find no statistically significant evidence that household or professional forecasts are systematically more accurate, either overall or within inflation regimes. Household expectation formation, by contrast, depends on the regime. When inflation becomes volatile, households put more weight on recently realized inflation, whereas neither professional survey shows a robust change in that weight. The rise in the household weight is robust to alternative standard errors, to most alternative regime definitions, and to uncertainty in the estimated regime. Our results support the joint monitoring of household and professional inflation forecasts by central banks, particularly in volatile periods.

Keywords: Inflation expectations; Survey forecasts; Markov-switching regimes; Forecast evaluation; Expectation formation; State dependence

JEL Classification: C22; C53; D84; E31; E37

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[*]Corresponding author. Email: thli@econ.sinica.edu.tw

1 Introduction

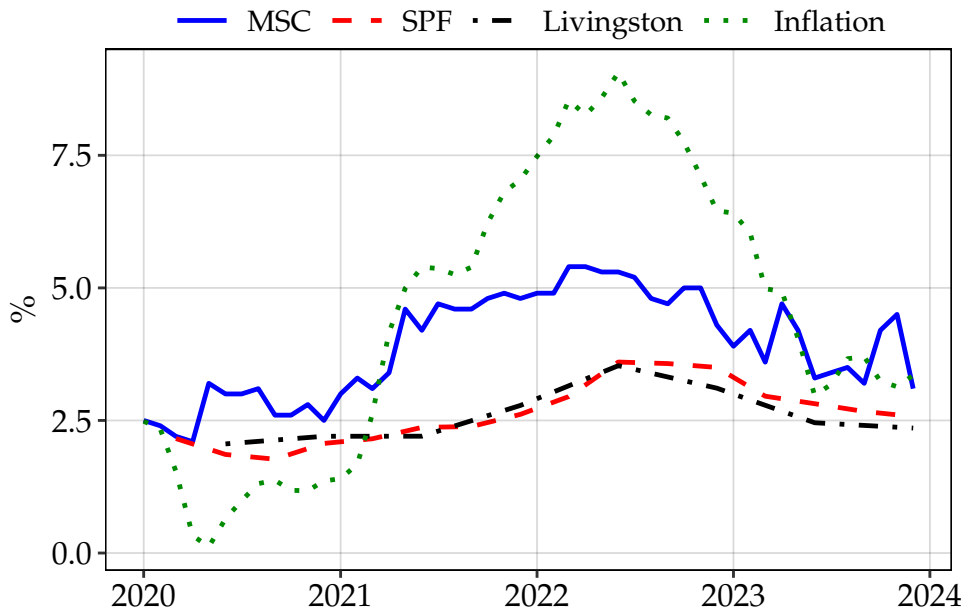
When prices in advanced economies rose abruptly in the early 2020s after three decades of relative stability, central banks faced a familiar question in an unfamiliar environment: what do people expect inflation to be? Expected inflation is embedded in the New Keynesian Phillips curve and monetary-policy rules, and it also deeply affects wage bargaining, durable goods purchases, and portfolio decisions (Bernanke 2007; Lieb and Schuffels 2022; Burke and Ozdagli 2023). Expectations, however, are unobservable. Applied work must therefore choose an empirical proxy, often between surveys of households and surveys of professional forecasters. That choice typically rests on a conventional characterization of the two groups, in which household inflation expectations are upward-biased, dispersed, and historically less accurate, while professional forecasts are treated as a more informed measure of expected inflation (Ang et al. 2007; D'Acunto et al. 2023; Weber et al. 2022; Henry et al. 2023).

The 2021–2023 episode is awkward for that characterization. Figure 1 plots median one-year-ahead inflation expectations from the University of Michigan Survey of Consumers (MSC), the Survey of Professional Forecasters (SPF), and the Livingston Survey against realized CPI inflation. As inflation accelerated, household expectations rose earlier and considerably further than professional forecasts. The MSC median reached 5.3% in June 2022, when inflation peaked near 9.1%, whereas the professional medians reached only about 3.6% in the SPF and 3.5% in the Livingston Survey. Despite their reputation for inattention, households moved first and furthest.

An episode by itself does not establish that households became better forecasters than professionals. The episode instead raises a different question: does the way each group forms its expectations change with the inflation environment? If it does, evidence on household and professional expectations from stable periods need not carry over to volatile ones.

Theories of costly information acquisition give a reason to expect such a change. In sticky-information models, agents update their information only occasionally (Mankiw and Reis 2002), and in rational-inattention models, they choose how much information to process given its cost (Sims 2003). When inflation is stable and the cost of ignoring it is small, households may devote little attention to aggregate price developments. When inflation begins moving sharply, the value of timely information rises. Empirical evidence is consistent with this view. Attention to inflation declined after the Great Inflation period (Pfäuti 2025), and household attention to inflation rises when inflation is high (Bracha and Tang 2025; Korenok et al. 2026; Weber et al. 2025). While these theories motivate our question, our analysis neither measures attention directly nor relies on it to interpret the results.

Figure 1: Expected and Realized CPI Inflation, 2020–2023



Notes: Median one-year-ahead inflation expectations of households from the monthly University of Michigan Survey of Consumers (MSC) and of professional forecasters from the quarterly Survey of Professional Forecasters (SPF) and the semiannual Livingston Survey, with realized year-over-year CPI inflation, 2020:M1–2023:M12. Sources: University of Michigan and U.S. Bureau of Labor Statistics, via the Federal Reserve Economic Data (FRED) database, and the Federal Reserve Bank of Philadelphia.

To answer this question, we examine how the formation of household and professional inflation expectations varies across inflation environments, and we revisit their relative accuracy. We compare the median one-year-ahead inflation expectations from the MSC, the SPF, and the Livingston Survey over 1981–2024 in stable and volatile inflation environments identified by a two-state Markov-switching autoregression fitted to realized CPI inflation (Hamilton 1989; Evans and Wachtel 1993). We date the regime at the forecast origin, so that it describes the inflation environment prevailing when expectations were formed. Within each regime, we examine forecast accuracy, unbiasedness, own-forecast efficiency, error persistence, the incorporation of public macroeconomic data, and the weight placed on recently realized inflation.

Our analyses yield three main findings. First, we find no statistically significant evidence that professional forecasts are systematically more accurate than household forecasts, or vice versa. Descriptively, professional forecasts exhibit lower root-mean-square error (RMSE) and mean absolute error (MAE) when inflation is stable, whereas household forecasts exhibit lower RMSE and MAE when inflation is volatile. Formal tests, however, do not establish either ranking. Across both professional comparison surveys, both loss functions, and both inflation regimes, none of the pairwise mean loss differences is statistically distinguishable from zero under Diebold and Mariano (1995) inference. Conditional predictive ability tests (Giacomini and White 2006) similarly show no evidence that the inflation regime or past relative forecast performance

predicts which survey will subsequently be more accurate. The fluctuation tests (Giacomini and Rossi 2010) reveal no subperiod in which one group achieves a statistically detectable advantage. Under squared loss, the mean loss differences stay insignificant in every robustness check of Section 6. All of these accuracy tests may have limited power because of overlapping one-year-ahead errors and few volatile-regime common dates, so we describe the two sets of forecasts as statistically comparable rather than demonstrably equal.

Second, our rationality tests reveal state dependence in forecast inefficiency. In volatile inflation environments, professional forecast errors become strongly predictable from the professionals' own forecasts. The own-forecast coefficients are -0.831 for the SPF and -0.871 for the Livingston Survey, both significant at the 1% level. This inefficiency survives every robustness check. Household forecasts show a similarly large point estimate in volatile periods. In contrast, the inefficiencies distinctive to households arise primarily in stable inflation environments, where household forecast errors show positive autocorrelation and are predictable from publicly available macroeconomic information. We read all three household patterns as suggestive. Thus, departures from rational expectations do not map cleanly onto a distinction between "uninformed households" and "informed professionals". Instead, the nature of forecast inefficiency itself varies across inflation environments and across forecaster types.

Third, and central to the paper, the weight households place on recently realized inflation depends on the inflation regime. In the stable regime, the cumulative loadings on recent inflation are 0.305 for the MSC, 0.284 for the SPF, and 0.302 for the Livingston Survey, each significant at the 1% level. In the volatile regime, the household loading rises to 0.383, a shift of 0.079 that is significant at the 1% level. The rise is robust to alternative standard errors, to most alternative regime definitions, to uncertainty in the estimated regime, and to ending the sample before the pandemic. Neither professional survey shows a robust shift.[1] In other words, when inflation turns volatile, households become more responsive to recent inflation, whereas professional forecasts show no robust change in their responsiveness.

Taken together, these results separate forecast accuracy from expectation formation. We cannot statistically rank household and professional forecasts by accuracy. We do find significant evidence that households raise the weight they place on recently realized inflation when inflation turns volatile, whereas we find no robust evidence that professionals do. The post-pandemic surge in Figure 1 fits this pattern. During that

[1]The Livingston shift of -0.109 is significant at the baseline, but its interval under the re-estimation of the regimes in Section 6.4 includes zero, and the shift itself turns positive when the intercept also varies by regime.

volatile period, households raised their inflation expectations earlier and further than professionals did.

Our paper makes three contributions. First, we compare how households and professionals form expectations across inflation regimes within one framework. The closest study is Gwak (2022), which uses MSC data through 2019 and a learning-from-experience model to show that households put more weight on recent inflation when inflation is high and volatile. We apply one distributed-lag regression to the MSC and to both professional surveys, so the two groups are compared on the same margin, and our sample covers the post-pandemic surge. We confirm the household pattern and find no robust counterpart for professionals. The inflation regime therefore shapes how households form their expectations, not only which forecast appears more accurate.

Second, we reassess the presumed professional advantage in inflation forecasting. Beyond conventional RMSE and MAE comparisons, we test for unconditional, conditional, and time-varying differences in predictive accuracy. Across these approaches, we cannot establish a professional accuracy advantage in either stable or volatile inflation regimes. Because neither group can be reliably ranked by forecast performance, the accuracy results motivate our focus on how each group forms its expectations.

Third, we treat the inflation regime as estimated rather than known. As in Gwak (2022), a Markov-switching model of inflation identifies the regimes, so we impose no *ex ante* threshold on the level of inflation. The two regimes differ above all in the variance of inflation shocks, and the volatile regime captures large movements in either direction, including rapid disinflation. Because the regime is estimated, we carry its uncertainty into the inference with a parametric bootstrap of the Markov-switching model, and we re-run the core results under six regime classifications, including filtered probabilities that remove the look-ahead of the smoothed baseline.

These findings have implications for how central banks assess inflation expectations. When inflation turns volatile, households put more weight on recently realized inflation, whereas professionals show no robust change in that weight. Neither group can be shown to forecast more accurately, so the accuracy evidence gives no reason to set household surveys aside in favor of professional forecasts. Reis (2021) argues that judging whether inflation expectations have become unanchored requires comparing several expectation measures, and our results add a reason to do so. More broadly, the choice of expectation measure in Phillips curves, monetary-policy rules, and other applications is a modeling assumption, not a choice that a presumed ranking of forecast accuracy can settle.

The empirical evidence of this paper relates to several strands of literature. A long

literature evaluates the accuracy and rationality of survey-based inflation expectations. Mehra (2002) finds that MSC expectations exhibit greater evidence of bias over 1961–2000, whereas Livingston Survey forecasts are broadly unbiased and efficient. Ang et al. (2007) show that survey forecasts generally outperform time-series and asset-market alternatives, with professional surveys modestly outperforming household expectations. Related research documents the upward bias, dispersion, and volatility of household inflation expectations (D'Acunto et al. 2023; Weber et al. 2022) and their historically weaker average forecast performance (Henry et al. 2023). Our evidence challenges this conventional ranking by showing that household and professional forecasts cannot be statistically ranked once the performance differences are formally tested, either unconditionally or conditional on the inflation environment.

Our analysis also relates to research showing that relative forecast performance can vary across economic environments. Diebold and Mariano (1995) provide the framework for testing unconditional differences in predictive accuracy, while Giacomini and White (2006) develop tests of conditional predictive ability and Giacomini and Rossi (2010) show how forecast rankings can change over time. Recent applications provide further evidence that relative forecast performance depends on economic conditions. Candelon and Roccazzella (2025), for example, document changes in relative inflation-forecast performance as macroeconomic uncertainty and financial conditions shifted. Related evidence also cautions against treating professional forecasts as a uniformly superior benchmark. El-Shagi (2011) shows that survey-based inflation expectations can perform comparably to sophisticated econometric forecasts, while Benchimol et al. (2022) document substantial heterogeneity in forecasting behavior and performance even among professional forecasters. Goodspeed (2025), most closely related to our accuracy comparison, finds that the professional advantage over households is concentrated in low-inflation periods. We condition on the volatility of inflation rather than its level, because periods in which inflation is difficult to track need not coincide with periods in which inflation is high.

Finally, our results relate to evidence that expectation formation is state dependent. Mankiw et al. (2003) provide the classic full-sample benchmark for the three surveys we use, finding that the median expectations in these surveys are consistent with neither full rationality nor pure adaptive expectations. Among households, Weber et al. (2025) show that exogenously provided inflation information has smaller effects when inflation is already elevated and respondents arrive better informed. Households also place disproportionate weight on salient and personally experienced prices (Malmendier and Nagel 2016; D'Acunto et al. 2021). Among professionals, information rigidity is state dependent (Coibion and Gorodnichenko 2015), and individual forecasters can overreact to news (Bordalo et al. 2020). Coibion and Gorodnichenko (2015),

Bordalo et al. (2020), and Pfäuti (2025) estimate how strongly forecasts respond to new information, given the previous forecast. Our adaptive regression instead measures the cumulative loading on twelve months of realized inflation and lets that loading differ between stable and volatile regimes.

The remainder of the paper proceeds as follows. Section 2 describes the survey data, forecast measures, and empirical methodology. Section 3 presents the inflation-regime classification. Section 4 evaluates relative forecast accuracy across regimes. Section 5 examines forecast rationality and expectation formation. Section 6 reports robustness checks. Section 7 concludes. Appendices A–E collect supplementary results.

2 Data, Forecast Measures, and Methodology

We use the median-based one-year-ahead expected rate of consumer price inflation from three long-running U.S. surveys: the monthly MSC for households, and the quarterly SPF and semiannual Livingston Survey for professionals. The sample starts in 1981, when the SPF begins asking about CPI inflation, and the evaluation window ends with 2024, the last calendar year of forecasts with realized one-year-ahead outcomes. Alongside the survey medians, we draw on realized CPI inflation as the forecast target, and on the three-month Treasury bill rate and the civilian unemployment rate, which enter the test of macroeconomic information and serve as controls in the adaptive regression. Appendix A collects the data details.

2.1 Survey measures of inflation expectations

The MSC has run since 1946, monthly since 1978, on a rotating panel of roughly 500 respondents. The survey asks whether prices in general will rise, fall, or stay the same over the next twelve months and "by about what percent". We use the published median of the point responses, the most widely used series and the only one available monthly over the full sample.[2]

The SPF, run by the American Statistical Association and the National Bureau of Economic Research (NBER) from 1968 and by the Federal Reserve Bank of Philadelphia since 1990, surveys roughly 30–40 professional forecasters each quarter. The SPF's one-year-ahead CPI question begins in 1981:Q3. The Livingston Survey began in 1946, and the Philadelphia Fed has likewise run it since 1990. The survey's roughly 20–30

[2]The New York Fed's Survey of Consumer Expectations elicits richer objects but begins only in June 2013, too short for regime-conditional analysis.

respondents are economists spanning academia, banking, government, and industry.

Realized inflation π_t is the year-over-year growth rate of the U.S. CPI for All Urban Consumers, not seasonally adjusted. Each survey yields a median-based expected inflation rate over the next twelve months. The target of a forecast formed at t is therefore π_{t+12} in monthly time, and the forecast error is

$$e_{t+h} \;=\; \pi_{t+h} - \mathbb{E}_t[\pi_{t+h}], \qquad h = 12 \text{ months}, \tag{1}$$

so a negative error means the survey over-predicted inflation and a positive error means it under-predicted inflation. We call t the *forecast origin* and $t + h$ the *realization date*.

We use two sample windows.

1. The evaluation window runs from the first common survey date through the last calendar year of forecasts with realized one-year-ahead outcomes: 1981:M9–2024:M12 for the MSC (520 months), 1981:Q3–2024:Q4 for the SPF (174 quarters), and 1981:H2–2024:H2 for the Livingston Survey (87 half-years).[3]

2. We estimate the Markov-switching model over 1981:Q2–2025:Q4. The first observation conditions the autoregression, so regime probabilities run from 1981:Q3, and we call 1981:Q3–2025:Q4 the classification window. The estimation sample extends one year past the evaluation window to include every realized value needed for the forecast-error evaluation.

2.2 A two-state Markov-switching model of inflation

We recover inflation regimes from the inflation process itself rather than imposing a threshold, estimating a two-state Markov-switching autoregression of order one for quarterly year-over-year CPI inflation:

$$\pi_t = \alpha_{s_t} + \rho_{s_t}\,\pi_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}\left(0, \sigma^2_{s_t}\right), \qquad s_t \in \{1, 2\}, \tag{2}$$

where the latent state s_t follows a first-order Markov chain with transition probabilities $p_{ij} = \Pr(s_t = j \mid s_{t-1} = i)$ and implied steady-state mean $\mu_s = \alpha_s/(1 - \rho_s)$. State-dependent intercepts and variances let the states differ in the level and variability of inflation. We impose two states for parsimony and interpretability.[4] We estimate the

[3] The October 2025 CPI was never published owing to the 2025 U.S. federal government shutdown, so MSC error samples lose the October 2024 forecast, leaving 519 rather than 520 months.

[4] For example, Evans and Wachtel (1993) model U.S. inflation as switching between two Markov regimes when decomposing inflation uncertainty, Gwak (2022) conditions household expectation for-

model by maximum likelihood with the expectation–maximization algorithm on the 1981:Q2–2025:Q4 sample above.

We ask how accurately households and professionals forecast inflation, and how each group forms its expectations, conditional on the inflation environment prevailing when the forecast is made. We characterize that environment *ex post* because the full sample gives the best available description of the conditions each forecaster faced. To this end, we define the regime dummy Regime_t as an indicator that the smoothed probability of state 2 exceeds one-half:

$$\text{Regime}_t \;=\; \mathbb{1}\{\Pr(s_t = 2 \mid \mathcal{I}_T) > 0.5\}, \tag{3}$$

where $\mathcal{I}_T$ is the full estimation sample. We date the regime dummy at the forecast origin t, the date at which the expectation is formed. Throughout, state 1 is the stable regime and state 2 the volatile regime.[5]

2.3 Evaluation specifications

We evaluate each survey over the full sample and within each regime on (1) accuracy, (2) four dimensions of rationality (unbiasedness, own-forecast efficiency, error persistence, and use of macroeconomic information), and (3) adaptiveness. Every specification interacts the object of interest with Regime_t. Baseline coefficients give stable-regime effects and baseline-plus-interaction sums give volatile-regime effects.

Accuracy. We measure accuracy by the RMSE and MAE of each survey's errors and, formally, by pairwise loss differences:

$$d_t = \mathcal{L}\left(e^{MSC}_{t+h}\right) - \mathcal{L}\left(e^{Prof}_{t+h}\right), \qquad \mathcal{L}(e) \in \{e^2,\ |e|\}, \tag{4}$$

where e^{MSC}_{t+h} and e^{Prof}_{t+h} are the household and professional survey errors on the one-year-ahead forecast made at t, so the loss difference is indexed by the forecast origin. We compute the loss differences on common dates, and a negative d_t indicates lower household loss relative to professionals.

In the spirit of Diebold and Mariano (1995), we test whether the mean loss differ-

mation on two such regimes, and Goodspeed (2025) evaluates household and professional forecast performance across two inflation states.

[5]The ordering is a normalization, because equation (2) identifies the states only up to relabeling. We fix the labels by calling the state with the higher innovation variance state 2.

ence is zero, overall and within each regime, with the regression:

$$d_t \;=\; \mu_S\,(1-\text{Regime}_t) + \mu_V\,\text{Regime}_t + \varepsilon_t, \tag{5}$$

with heteroskedasticity- and autocorrelation-consistent (HAC) inference on μ_S and μ_V and moving-block bootstrap (MBB) intervals as a complement. Imposing $\mu_S = \mu_V$ (a single constant in place of the two regime indicators) gives the full-sample test of $\mathbb{E}[d_t] = 0$.

Rationality. The four rationality tests share one benchmark. Under full-information rational expectations, forecast errors are unbiased and orthogonal to the period-t information set. The specifications follow Mankiw et al. (2003), and we extend them with the regime interactions.

Unbiasedness. We regress the error on a constant and the regime dummy:

$$e_{t+h} = \alpha_1 + \alpha_2 \cdot \text{Regime}_t + \varepsilon_{t+h}, \tag{6}$$

so α_1 is the mean error in the stable regime and $\alpha_1 + \alpha_2$ the mean in the volatile regime. Under unbiasedness, both means are zero, and a negative mean indicates systematic over-prediction.

Own-forecast efficiency. We add the forecast itself:

$$e_{t+h} = \alpha_1 + \beta_1 \mathbb{E}_t[\pi_{t+h}] + (\alpha_2 + \beta_2 \mathbb{E}_t[\pi_{t+h}]) \cdot \text{Regime}_t + \varepsilon_{t+h}. \tag{7}$$

A nonzero β_1, or volatile-regime sum $\beta_1 + \beta_2$, signals unexploited information.

Error persistence. We replace the forecast with the survey's most recent own error, $e_t = \pi_t - \mathbb{E}_{t-12}[\pi_t]$:

$$e_{t+h} = \alpha_1 + \beta_1 e_t + (\alpha_2 + \beta_2 e_t) \cdot \text{Regime}_t + \varepsilon_{t+h}. \tag{8}$$

A nonzero β_1 or sum $\beta_1 + \beta_2$ means past errors predict the current one.

Use of macroeconomic information. We extend equation (7) with lagged inflation π_{t-1}, the lagged three-month Treasury bill rate i_{t-1}, and the lagged civilian unemployment rate $\mathcal{U}_{t-1}$. The bill rate and the unemployment rate are available monthly in real time and enter lagged one month, for all three surveys, so they sit unambiguously in the information set at the forecast origin:

$$\begin{aligned} e_{t+h} = {} & \alpha_1 + \beta_1 \mathbb{E}_t[\pi_{t+h}] + \gamma_1 \pi_{t-1} + \kappa_1 i_{t-1} + \delta_1 \mathcal{U}_{t-1} \\ & + (\alpha_2 + \beta_2 \mathbb{E}_t[\pi_{t+h}] + \gamma_2 \pi_{t-1} + \kappa_2 i_{t-1} + \delta_2 \mathcal{U}_{t-1}) \cdot \text{Regime}_t + \varepsilon_{t+h}. \end{aligned} \tag{9}$$

We focus on the joint use of macroeconomic information. A rational forecast embeds these public series, so γ_1, κ_1, and δ_1 should be jointly zero. The volatile-regime sums $\gamma_1 + \gamma_2$, $\kappa_1 + \kappa_2$, and $\delta_1 + \delta_2$ should likewise be jointly zero.

Expectation adaptiveness. We regress the survey forecast on the twelve monthly lags of realized inflation preceding the forecast origin, with regime-varying coefficients, controlling for current and lagged unemployment and interest rates:

$$\mathbb{E}_t[\pi_{t+h}] = \alpha + \big(\beta_1(L) + \beta_2(L) \cdot \text{Regime}_t\big)\, \pi_t + \lambda_1\, \mathcal{U}_t + \lambda_2\, \mathcal{U}_{t-1} + \theta_1\, i_t + \theta_2\, i_{t-1} + \varepsilon_t, \tag{10}$$

where L is the lag operator and $\beta_k(L) = \sum_{j=1}^{12} \beta_{kj} L^j$ for $k = 1, 2$. Our interest is in the sum of the lag coefficients, not the individual terms. We call $\beta_k(1) = \sum_{j=1}^{12} \beta_{kj}$ the *cumulative loading*. It measures the total change in the forecast per percentage point of realized inflation, once all twelve lags have acted. The cumulative loading is $\beta_1(1)$ in the stable regime and $\beta_1(1) + \beta_2(1)$ in the volatile regime.

Inference. Forecast errors here are serially correlated by construction, because successive one-year forecast windows overlap. All standard errors in the survey evaluations therefore come from *prewhitened* Newey–West estimators (Newey and West 1987; Andrews and Monahan 1992) with a bandwidth of one year at each survey's frequency (twelve, four, and two lags, so every bandwidth covers the overlap). We complement these standard errors with MBB intervals and p-values based on one-year blocks and detail the resampling mechanics in Appendix E.3.

3 Inflation Regime Identification

Table 1 reports the estimates of the Markov-switching model of Section 2.2. Regime 1 combines a steady-state mean of $\widehat{\mu}_1 = 2.212$ with a small innovation variance of 0.211. Regime 2 combines a mean of $\widehat{\mu}_2 = 3.693$ with an innovation variance of 1.573, about seven and a half times larger. Both states are highly persistent, with continuation probabilities $\widehat{p}_{11} = 0.950$ and $\widehat{p}_{22} = 0.899$ and expected durations of about five years and about two and a half years. We name the states *stable* and *volatile*, not low- and high-inflation, because the variance gap is what separates them. The volatile state captures large inflation movements, rising or falling, rather than only "high inflation".

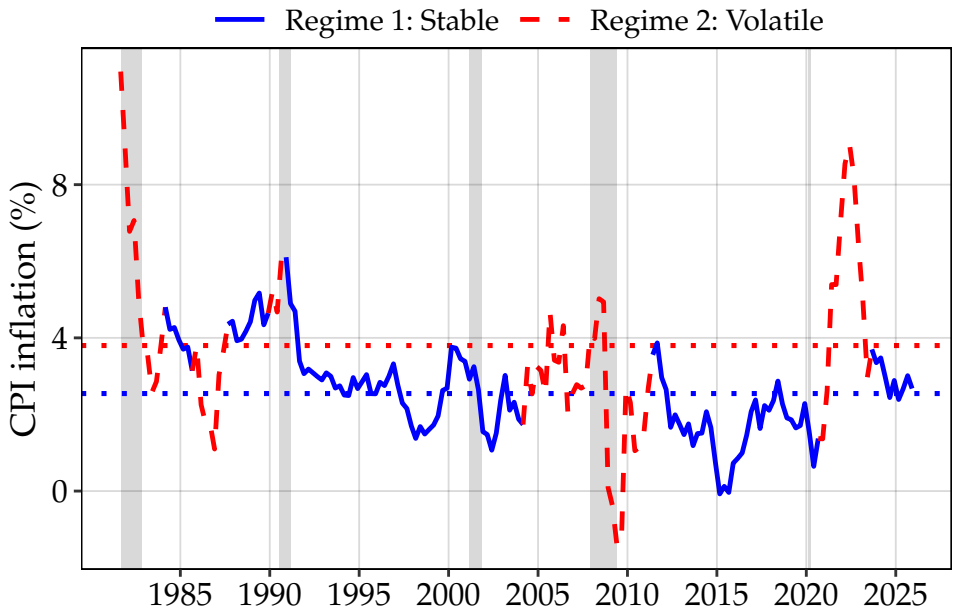
Figure 2 displays realized inflation colored by the classification of equation (3). Five volatile episodes cover 64 of the 178 quarters, with nine regime transitions. The episodes are the early-1980s disinflation (1981:Q3–1984:Q1), 1985:Q4–1987:Q3,

Table 1: Estimated Two-State Markov-Switching Model of CPI Inflation

	$\widehat{\alpha}_s$	$\widehat{\rho}_s$	$\widehat{\sigma}_s^2$	$\widehat{\mu}_s$	$\widehat{p}_{ss}$
Regime 1: Stable	0.282	0.873	0.211	2.212	0.950
Regime 2: Volatile	0.713	0.807	1.573	3.693	0.899

Notes: Maximum-likelihood estimates of the Markov-switching first-order autoregression, MS-AR(1), in equation (2), fitted to quarterly year-over-year CPI inflation, measured in the final month of each quarter, over the 178 quarters of 1981:Q3–2025:Q4, with the 1981:Q2 observation serving as the conditioning value. $\widehat{\alpha}_s$, $\widehat{\rho}_s$, and $\widehat{\sigma}_s^2$ are the intercept, autoregressive coefficient, and innovation variance in regime s; $\widehat{\mu}_s = \widehat{\alpha}_s/(1-\widehat{\rho}_s)$ is the implied steady-state mean of inflation, in percent, and $\widehat{p}_{ss} = \Pr(s_{t+1} = s \mid s_t = s)$ the estimated probability of remaining in regime s next quarter. Standard errors of the intercept and autoregressive parameters: 0.087 and 0.031 (Regime 1), 0.173 and 0.038 (Regime 2).

Figure 2: Realized Inflation Classified by Inflation Regime



Notes: Quarterly year-over-year CPI inflation, measured in the final month of each quarter, 1981:Q3–2025:Q4. A quarter is classified as volatile when the smoothed probability of the volatile state exceeds 0.5. The two horizontal dotted lines mark the mean of realized inflation within each regime, in the matching color. Shaded bars denote NBER recessions. Sources: U.S. Bureau of Labor Statistics, via FRED, and authors' estimates of equation (2).

1990:Q1–1990:Q4, 2004:Q2–2011:Q2 (the pre-crisis commodity run-up through the financial crisis and the European sovereign-debt period), and the post-pandemic surge (2020:Q4–2023:Q3). The indicator Regime_t measures how hard inflation is to track, not only how high it stands. The figure also shows what a level classification would miss. A threshold near 3% would label many volatile quarters as low-inflation periods, such as the years after the financial crisis.[6]

Tests that do not rely on the Markov-switching model support reading the inflation environment off the variance. Most clearly on monthly data, they find that the instability of inflation lies in the variance of its innovations rather than in its conditional mean.

[6]Of the 64 volatile quarters, 23 have inflation below the evaluation-window median of 2.8%, and a 4% level rule recovers only 24 of the 64. The standard deviation of the quarterly change in inflation, by contrast, is 1.38 percentage points in the volatile regime against 0.47 in the stable one.

The regime label, in turn, predicts the size of inflation innovations, including through the post-2021 surge. Appendix B reports these checks.[7]

4 Is There a Professional Accuracy Advantage?

This section evaluates whether professional forecasters have a systematic accuracy advantage over households. We find no statistically significant evidence of an accuracy advantage for either group. The descriptive edges of professionals in the stable regime and of households in the volatile regime both dissolve under formal testing. Section 4.1 documents the pattern in the forecast errors, and Section 4.2 presents formal pairwise tests of equal predictive accuracy. Section 4.3 extends the analysis with a conditional predictive ability test and a window-by-window fluctuation test. Section 4.4 explains why neither edge survives testing and what that implies for the choice among surveys. In Appendix C, we specify the two tests of Section 4.3 and probe the no-advantage answer for significance levels and direction.

4.1 Descriptive accuracy across regimes

Table 2 reports the RMSE and MAE of one-year-ahead forecast errors for each survey. Over the full sample, the three surveys are close to indistinguishable. The MSC attains a slightly lower RMSE and the professional surveys a slightly lower MAE, so no survey dominates on both criteria.

The regime decomposition reveals a descriptive pattern that the full sample hides. In the stable regime, the professional surveys are more accurate on both criteria, with a household RMSE of 1.21 against 1.04 for the SPF and 1.01 for Livingston. In the volatile regime, the point estimates reverse on both criteria, with a household RMSE of 2.08 against 2.26 for both professional surveys. The professional stable-regime edge is 0.17–0.20 RMSE points, roughly 15% of the household RMSE. The household volatile-regime edge is 0.17–0.18 RMSE points, roughly 8% of the professional RMSE. Whether either edge is distinguishable from zero, RMSE tables cannot say.

[7] The parameter-constancy test of Nyblom (1989) and Hansen (1992) locates the instability in the innovation variance, not the conditional mean, most sharply at monthly frequency. The cumulative-sum-of-squares test of Inclán and Tiao (1994), corrected for conditional heteroskedasticity as in Sansó et al. (2004), rejects a constant variance at the 1% level on monthly data, so the variance regime is not merely generalized autoregressive conditional heteroskedasticity (GARCH). An iterative nonparametric search confirms the timing, placing its two change points three quarters inside the transitions that open and close the model's longest volatile episode. Re-estimating the model on core inflation, which excludes food and energy, preserves the early-1980s and post-pandemic episodes at the two ends of the sample, so energy accounts for the volatile episodes in the middle of the sample rather than those at its ends.

Table 2: Forecast Accuracy by Inflation Regime

		MSC	SPF	Livingston
Full sample	RMSE	1.586	1.597	1.589
	MAE	1.168	1.085	1.093
Stable regime	RMSE	1.208	1.037	1.010
	MAE	0.932	0.784	0.780
Volatile regime	RMSE	2.085	2.256	2.261
	MAE	1.576	1.603	1.632

Notes: Root-mean-square error (RMSE) and mean absolute error (MAE) of the one-year-ahead forecast errors of equation (1), in percentage points. Samples: MSC 519 monthly (1981:M9–2024:M12), SPF 174 quarterly, Livingston 87 semiannual; the October 2024 MSC forecast is excluded because the October 2025 CPI was never published. Regimes follow Regime_t of equation (3), dated at the forecast origin. The stable regime contains 329 MSC months, 110 SPF quarters, and 55 Livingston half-years, and the volatile regime 190, 64, and 32.

4.2 Tests of equal predictive accuracy

We test whether the mean of the pairwise loss difference d_t of equation (4) is zero, overall and within each regime, following equation (5). Table 3 reports twelve cells. Its rows cross the squared and absolute loss functions with the MSC–SPF and MSC–Livingston comparisons, and its columns cover the full sample and the stable and volatile regimes. Each cell reports the mean loss difference, its prewhitened Newey–West standard error in parentheses, and a moving-block bootstrap 95% interval in brackets. Our decision rule is conservative and symmetric. One survey outperforms the other only if an interval lies entirely on one side of zero. Otherwise the two are *comparable* in that regime.

For both professional comparisons and both loss functions, the point estimates track the descriptive pattern, positive in the stable regime and negative in the volatile regime. None of the twelve mean differences is statistically distinguishable from zero. The largest point estimate in absolute value, the MSC–SPF squared-loss difference of -0.592 in the volatile regime, carries a standard error of 2.185 and a bootstrap interval of $[-3.120,\ 2.233]$.

4.3 Conditional and window-by-window accuracy tests

The tests of Section 4.2 compare regime means, and a mean can conceal two failures of equal accuracy. First, relative performance may be predictable from information available at the origin, in which case a zero mean hides a usable switching rule. Second, one survey may dominate over a subperiod too short to move the mean. We take the two in turn, with the conditional predictive ability test of Giacomini and White (2006) and the fluctuation test of Giacomini and Rossi (2010). Appendix C specifies both, in

Table 3: Pairwise Loss-Difference Tests by Inflation Regime

Comparison	Full sample	Stable regime	Volatile regime
MSC − SPF, squared loss	0.038 (0.877) [−0.962, 1.137]	0.405 (0.410) [−0.173, 0.993]	−0.592 (2.185) [−3.120, 2.233]
MSC − SPF, absolute loss	0.088 (0.130) [−0.099, 0.304]	0.163 (0.140) [−0.041, 0.370]	−0.040 (0.233) [−0.406, 0.381]
MSC − Livingston, squared loss	0.009 (0.628) [−1.035, 1.168]	0.357 (0.321) [−0.131, 0.885]	−0.588 (1.538) [−3.327, 2.397]
MSC − Livingston, absolute loss	0.083 (0.117) [−0.105, 0.299]	0.161 (0.141) [−0.022, 0.356]	−0.050 (0.201) [−0.444, 0.401]

Notes: Entries are mean loss differences $\bar{d}$ of equation (4), estimated by equation (5), in percentage points (squared under squared loss); negative entries indicate lower loss for the MSC. Prewhitened Newey–West standard errors in parentheses (bandwidth of one year: four lags for MSC–SPF, two for MSC–Livingston) and moving-block bootstrap 95% percentile intervals in brackets (9,999 replications, resampling d_t and Regime_t jointly in one-year blocks). Regimes follow Regime_t of equation (3), dated at the forecast origin. Common dates: 174 quarterly observations (110 stable, 64 volatile) for MSC–SPF and 87 semiannual observations (55 stable, 32 volatile) for MSC–Livingston. No entry is significantly different from zero at the 10% level under Newey–West inference.

equations (C.1) and (C.2).

Does anything known at the origin predict relative accuracy? The conditional test regresses the loss difference on instruments dated at the forecast origin and asks whether they jointly predict it. Those instruments must be measurable with respect to the origin's information set, and that requirement bites on our own regime indicator. The baseline of equation (3) is built from smoothed probabilities, which condition on the full sample, so it is not measurable at the origin. We therefore run every test twice, once with the baseline indicator and once with the filtered indicator $\text{Regime}_{f,t}$, which conditions only on inflation realized through t. Across the sixteen tests of Table 4, no Wald statistic rejects. The smallest Newey–West p-value is 0.140 and the smallest bootstrap p-value 0.255.

With the regime alone as the instrument, making the regime indicator measurable moves the tests away from rejection, not toward it. Replacing the baseline indicator with the filtered one raises the p-value from 0.580 to 0.927 for the MSC–SPF squared-loss comparison and from 0.483 to 0.915 for MSC–Livingston, and it raises both absolute-loss p-values as well. The reason is that filtering relabels about one date in eight, and the relabeled dates track the loss difference no better than noise. When the most recent observable loss difference d_{t-s} joins the regime as an instrument, the filtered indicator gives slightly smaller p-values than the baseline one, but none falls below 0.140. The no-advantage answer therefore does not rest on the look-ahead in the

Table 4: Giacomini–White Conditional Predictive Ability Tests

	MSC vs. SPF			MSC vs. Livingston		
Instruments	W	p_{NW}	p_{MBB}	W	p_{NW}	p_{MBB}
Panel A: squared loss						
$\{1,\ \text{Regime}_t\}$	1.091	0.580	0.580	1.456	0.483	0.559
$\{1,\ \text{Regime}_t,\ d_{t-s}\}$	1.055	0.788	0.809	2.460	0.483	0.576
$\{1,\ \text{Regime}_{f,t}\}$	0.151	0.927	0.932	0.177	0.915	0.935
$\{1,\ \text{Regime}_{f,t},\ d_{t-s}\}$	1.243	0.743	0.770	2.869	0.412	0.505
Panel B: absolute loss						
$\{1,\ \text{Regime}_t\}$	1.443	0.486	0.502	1.416	0.493	0.519
$\{1,\ \text{Regime}_t,\ d_{t-s}\}$	2.733	0.435	0.501	4.499	0.212	0.330
$\{1,\ \text{Regime}_{f,t}\}$	0.896	0.639	0.660	0.740	0.691	0.717
$\{1,\ \text{Regime}_{f,t},\ d_{t-s}\}$	3.099	0.377	0.438	5.484	0.140	0.255

Notes: Wald tests of $\mathbb{E}[d_t \mid \mathcal{F}_t] = 0$, where $\mathcal{F}_t$ is the information set at the forecast origin (Giacomini and White 2006). W is the Wald statistic of equation (C.1) for the null that all coefficients are zero in a regression of the loss difference d_t, the MSC loss minus the professional loss, on the instruments. The covariance is prewhitened Newey–West with a bandwidth of one year, four lags quarterly and two semiannual, and p_{NW} refers W to its asymptotic χ^2 distribution, with degrees of freedom equal to the number of instruments, two or three. p_{MBB} is a recentered moving-block bootstrap p-value from 4,999 replications, resampling d_t and the instruments jointly in one-year blocks. Regime_t is the baseline indicator of equation (3), built from smoothed probabilities. $\text{Regime}_{f,t}$ replaces those with filtered probabilities, which condition only on inflation realized through t. It is measurable at the forecast origin. The instrument d_{t-s} is the most recent loss difference observable at the origin, with $s = 5$ quarters or three half-years; including it cuts n from 174 to 169 quarterly and from 87 to 84 semiannual.

classification, since even the indicator that draws on the full sample predicts nothing.[8]

Is the no-advantage answer an average over offsetting episodes? The fluctuation test computes the loss difference over rolling windows and compares the largest standardized value along the path with a critical value for the path as a whole. The MSC–SPF squared-loss path rises as far as 1.78 while the windows fill with the stable 2010s, favoring the professionals, then falls through zero as the volatile quarters of 2021–2023 enter. It never approaches its critical values, 3.01 at the longer window and 3.18 at the shorter, in either direction, and the MSC–Livingston path behaves the same way. Under absolute loss, the paths climb higher, peaking in windows dominated by the stable 2010s, and come within about half a point of their critical values without crossing them. Figure C.1 in Appendix C plots all eight paths, and the appendix reports the statistics. No window in more than four decades delivers a ranking that the full-sample mean conceals.

[8]The filtered indicator is still not fully real time, because its probabilities condition only on inflation realized through t while the model parameters behind them come from the full sample. A fully recursive construction, re-estimating those parameters at every origin, would add estimation noise to the relabeling noise, and the regime-only comparison in the text shows that such noise pushes the p-values up, not down. Dating the filtered indicator one quarter before the origin, which removes even the origin quarter's information, likewise changes no conclusion. The smallest Newey–West p-value across the eight tests is then 0.210 and the smallest bootstrap p-value 0.316.

4.4 Discussion

The descriptive gaps in Table 2 dissolve under formal testing for three reasons. First, the gaps are small. No regime advantage exceeds 0.20 RMSE points against error levels between 1.01 and 2.26. Second, overlapping twelve-month forecast windows make the errors serially correlated. That persistence cuts the effective size of the stable and volatile quarterly samples by factors of roughly five and six under squared loss.[9] Third, the performance reversal occurs in the noisiest cell of the design. Under squared loss, differences scale with error magnitude, and the volatile regime makes the errors large precisely where the sample size is smaller. These three are features of the data rather than artifacts of the estimator. No conclusion in Table 3 differs between the Newey–West standard errors and the bootstrap intervals.

The 95% bootstrap intervals in Table 3 bound the size of any accuracy edge the tests might have missed, measured against the professional surveys' mean loss in Table 2. In the stable regime, the intervals are tight. The largest household advantage consistent with the data is 5% of the SPF's mean absolute error and 16% of its mean squared error, with corresponding bounds of 3% and 13% against the Livingston Survey. In the volatile regime, however, the intervals are wide, admitting household advantages of 25% and 61% of the SPF's mean absolute and mean squared error, and 27% and 65% against the Livingston Survey.

The limited directional support in the sample favors the professional advantage, and only within the stable regime. In the one-sided bootstrap tail areas of Appendix C, the professional edge in quiet times draws some directional support, while the household edge in turbulent times draws none. This asymmetry has a practical consequence. Quiet times are precisely when the choice among surveys carries the lowest stakes because errors are small for all three surveys. Turbulent times, when inflation is moving rapidly and expectations are most consequential for policy, are exactly when the choice carries the highest stakes. The presumed professional advantage is thus weakest where a reliable benchmark is needed most.

Because neither group is demonstrably more accurate, the choice of expectations proxy must be argued rather than presumed. The main question is then not a performance horse race but whether the way each group forms expectations changes with the inflation environment.

[9] The quarterly comparison rests on 110 stable and 64 volatile common dates, and the effective sample sizes fall to 23 and 10, as reported in Table E.1.

5 Does Expectation Formation Depend on the Inflation Regime?

This section documents two results on expectation formation across inflation regimes. First, the departures from rational expectations that are specific to households occur when inflation is stable, and we read them as suggestive. In the volatile regime, own-forecast inefficiency is robust for the professionals and suggestive for households. Second, when the environment turns volatile, households raise the weight they place on recently realized inflation, whereas neither professional survey shows a robust shift in that weight.

Section 5.1 runs the four rationality tests behind the first result, Section 5.2 estimates the adaptive-expectations regressions behind the second, and Section 5.3 discusses what the two results mean. Appendix D collects the supporting tables.

5.1 Four dimensions of rationality

Unbiasedness. No survey shows significant bias in either regime in Table 5, where we estimate equation (6). The stable-regime point estimates of α_1 are negative for all three surveys, at -0.429 for the MSC, -0.273 for the SPF, and -0.260 for Livingston. The volatile-regime sums are small and of mixed sign. We read the negative point estimates as consistent with mild stable-regime over-prediction for both groups, which the data cannot establish.

Table 5: Tests of Forecast Bias across Regimes

	MSC	SPF	Livingston
α_1	-0.429	-0.273	-0.260
	(0.264)	(0.219)	(0.173)
	$[-0.878,\ 0.044]$	$[-0.710,\ 0.258]$	$[-0.731,\ 0.290]$
α_2	0.335	0.505	0.528
	(0.780)	(0.686)	(0.554)
	$[-0.417,\ 1.083]$	$[-0.221,\ 1.286]$	$[-0.286,\ 1.372]$
$\alpha_1 + \alpha_2$	-0.094	0.232	0.268
	(0.811)	(0.736)	(0.597)
	$[-0.677,\ 0.509]$	$[-0.334,\ 0.906]$	$[-0.339,\ 0.968]$

Notes: Estimates of equation (6), a regression of the one-year-ahead forecast error, realized minus expected inflation in percentage points, on a constant and Regime$_t$ of equation (3), which equals one when the smoothed probability of the volatile regime at the forecast origin exceeds 0.5. Subscript 1 marks stable-regime coefficients, subscript 2 volatile-regime shifts, and their sums volatile-regime values. Prewhitened Newey–West standard errors in parentheses. Brackets: residual moving-block bootstrap 95% percentile intervals from 4,999 replications. Samples: MSC 519 monthly (1981:M9–2024:M12 less October 2024, as the October 2025 CPI was never published), SPF 174 quarterly, Livingston 87 semiannual.

Table 6: Tests of Own-Forecast Efficiency across Regimes

	MSC	SPF	Livingston
β_1	-0.208 (0.479) $[-1.018,\ 0.614]$	-0.230 (0.272) $[-0.681,\ 0.274]$	-0.173 (0.222) $[-0.641,\ 0.335]$
β_2	-0.706 (0.809) $[-1.672,\ 0.270]$	-0.601^{*} (0.345) $[-1.196,\ -0.030]$	-0.698^{**} (0.289) $[-1.324,\ -0.064]$
$\beta_1 + \beta_2$	-0.913 (0.736) $[-1.461,\ -0.341]$	-0.831^{***} (0.288) $[-1.223,\ -0.415]$	-0.871^{***} (0.245) $[-1.285,\ -0.399]$
α_1	0.179 (1.215) $[-2.275,\ 2.588]$	0.368 (0.809) $[-1.055,\ 1.774]$	0.228 (0.634) $[-1.250,\ 1.647]$
α_2	3.014 (2.157) $[-0.185,\ 6.101]$	2.565^{*} (1.543) $[0.682,\ 4.490]$	2.742^{**} (1.299) $[0.707,\ 4.796]$
$\alpha_1 + \alpha_2$	3.193^{*} (1.897) $[1.104,\ 5.232]$	2.933^{*} (1.538) $[1.526,\ 4.419]$	2.969^{**} (1.300) $[1.506,\ 4.508]$
L1: Reject H_0: $\alpha_1 = \beta_1 = 0$?	F=1.322 p=0.268 p_{MBB}=0.526	F=1.287 p=0.279 p_{MBB}=0.503	F=1.376 p=0.258 p_{MBB}=0.512
L2: Reject H_0: $\alpha_1 + \alpha_2 = \beta_1 + \beta_2 = 0$?	F=2.062 p=0.128 p_{MBB}=0.230	F=17.103 p<0.001 p_{MBB}=0.003	F=29.012 p<0.001 p_{MBB}=0.001

Notes: Estimates of equation (7), which adds the survey forecast to the regression of Table 5; β is its coefficient, and regimes and subscripts are as there. Prewhitened Newey–West standard errors in parentheses. Brackets: residual moving-block bootstrap (MBB) 95% percentile intervals from 4,999 replications. L1 and L2: prewhitened Newey–West F statistic and its p-value, and null-imposed residual-MBB p-value p_{MBB} from 1,999 replications. Samples as in Table 5. Stars from Newey–West t-tests: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Own-forecast efficiency. Under informational efficiency the forecast error should be orthogonal to the forecast itself. Table 6, where we estimate equation (7), establishes own-forecast inefficiency for the professionals in the volatile regime. For households, the volatile-regime point estimate is as large, but the evidence is only suggestive.

For the professionals, β_1 is insignificant in the stable regime for both surveys. In the volatile regime, the sum $\beta_1 + \beta_2$ is significantly negative, at -0.831 for the SPF and -0.871 for Livingston, both at the 1% level, and the joint test rejects overwhelmingly, with F statistics of 17.1 and 29.0. A coefficient of this size is economically large. In volatile times, a forecast one percentage point higher comes with over-prediction more than eight tenths of a point larger, predictable from the forecast alone.

For households, the stable-regime coefficient of -0.208 and the volatile-regime sum of -0.913 are both insignificant under the prewhitened standard errors, and neither joint test rejects. The bootstrap interval for the volatile-regime sum excludes zero, but the null-imposed joint bootstrap does not reject, so we read the household evidence as suggestive rather than established.

Table 7: Tests of Forecast-Error Persistence across Regimes

	MSC	SPF	Livingston
β_1	0.334^{**}	0.120	0.169
	(0.168)	(0.181)	(0.203)
	$[-0.050, 0.707]$	$[-0.394, 0.644]$	$[-0.459, 0.805]$
β_2	-0.422	0.027	0.018
	(0.317)	(0.293)	(0.283)
	$[-0.857, 0.020]$	$[-0.537, 0.588]$	$[-0.661, 0.704]$
$\beta_1 + \beta_2$	-0.088	0.147	0.186
	(0.288)	(0.214)	(0.173)
	$[-0.334, 0.156]$	$[-0.082, 0.385]$	$[-0.068, 0.450]$
α_1	-0.270	-0.231	-0.204
	(0.211)	(0.189)	(0.158)
	$[-0.741, 0.239]$	$[-0.662, 0.279]$	$[-0.692, 0.358]$
α_2	0.176	0.643	0.417
	(0.713)	(0.464)	(0.422)
	$[-0.558, 0.906]$	$[-0.120, 1.425]$	$[-0.384, 1.257]$
$\alpha_1 + \alpha_2$	-0.094	0.411	0.213
	(0.748)	(0.479)	(0.418)
	$[-0.659, 0.489]$	$[-0.127, 1.091]$	$[-0.365, 0.900]$

Notes: Estimates of equation (8), which adds the error of the survey's forecast made twelve months earlier to the regression of Table 5; β is its coefficient, and regimes and subscripts are as there. Prewhitened Newey–West standard errors in parentheses. Brackets: residual moving-block bootstrap 95% percentile intervals from 4,999 replications. Samples as in Table 5, except that the SPF loses its first four quarters for lack of an earlier forecast, leaving 170. Stars from Newey–West t-tests: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Error persistence. Only household errors fail equation (8), and only in the stable regime, as reported in Table 7. When inflation is stable, household errors show positive persistence, with $\beta_1 = 0.334$ significant at the 5% level. The moving-block bootstrap interval of $[-0.050, 0.707]$ narrowly includes zero, so we read the persistence as suggestive rather than established. Professional errors show no persistence in either regime.

Use of macroeconomic information. The two groups fail the macroeconomic-information restriction in mirror image. Household errors are predictable from public macroeconomic data when inflation is stable, whereas professional errors are predictable when it is volatile. Rationality requires that lagged inflation, the short-term interest rate, and the unemployment rate not predict the forecast error. We estimate equation (9) in Table 8, with L3 testing the stable-regime coefficients and L4 their volatile-regime sums.

For households, L3 rejects at the 5% level, with $F = 2.732$ and $p = 0.043$, while the volatile-regime test does not reject. The interest-rate coefficient $\kappa_1 = 0.206$ is positive and significant only at the 10% level, so inflation later runs above the household forecast when interest rates are high, which points to under-reaction to interest-rate

news.[10] For the professionals, the pattern runs the other way. L3 does not reject, while L4 rejects at the 5% level for the SPF, with $p = 0.038$, and only at the 10% level for Livingston, with $p = 0.078$. The volatile-regime unemployment sum $\delta_1 + \delta_2$ is significantly negative for both.

The null-imposed moving-block bootstrap qualifies both halves of the pattern. It confirms the household stable-regime rejection only at the 10% level, with a p-value of 0.089, and it does not confirm the professional volatile-regime rejections, with p-values of 0.214 for the SPF and 0.390 for Livingston. We therefore read both halves as suggestive rather than established.

5.2 Expectation adaptiveness

In Table 9, we estimate the distributed-lag regression of equation (10) and report the cumulative loadings defined there, $\beta_1(1)$ for the stable regime and $\beta_1(1) + \beta_2(1)$ for the volatile regime.[11] Three results follow.

First, in the stable regime all three surveys put significant weight on recently realized inflation. The cumulative loadings are 0.305 for the MSC, 0.284 for the SPF, and 0.302 for Livingston, each significant at the 1% level.

Second, in the volatile regime the household weight rises, whereas the professional weights show no robust change. The regime interaction $\beta_2(1)$ is positive and significant for households, at 0.079 with $p < 0.01$.[12] For the SPF, it is negative though insignificant, at -0.042, and for Livingston it is negative and significant, at -0.109 with $p < 0.01$. The volatile-regime loading thus rises to 0.383 for households, while the professional point estimates fall to 0.242 for the SPF and 0.193 for Livingston. The residual moving-block bootstrap agrees survey by survey, with intervals of $[0.033,\ 0.125]$ for the MSC, $[-0.102,\ 0.019]$ for the SPF, and $[-0.175,\ -0.040]$ for Livingston. The single-lag specification of Table D.3 in Appendix D shows the same sign pattern. The Livingston fall is less secure, because it reverses sign once the intercept also varies by regime, as

[10] The MSC stable-regime β_1 of -1.026 in Table 8, significant at the 1% level, does not contradict the insignificant -0.208 of Table 6. Conditioning on π_{t-1}, i_{t-1}, and $\mathcal{U}_{t-1}$ makes β_1 the coefficient on the part of the forecast orthogonal to those variables, so the joint L3 test is the better guide.

[11] The adaptive regression measures one margin of expectation formation. Following Pfajfar and Santoro (2010), we also regress the forecast revision on the most recent verifiable forecast error, which measures how fast each survey corrects its own mistakes. The regime interaction is insignificant for all three surveys in Table D.4 of Appendix D, so we cannot establish a regime shift on that margin.

[12] The household shift does not rest on the post-pandemic surge. Holding the regime classification fixed, ending the sample in 2019:M12 gives a shift of 0.076, and dropping the 2020:Q4–2023:Q3 episode gives 0.073, both significant at the 5% level with residual moving-block bootstrap intervals above zero. In both samples, the SPF shift stays insignificant and the Livingston shift stays negative and significant, at -0.127 and -0.118.

Table 8: Tests of the Use of Macroeconomic Information across Regimes

	MSC	SPF	Livingston
α_1	0.710	0.283	−0.058
	(1.085)	(0.944)	(0.908)
	[−1.696, 3.120]	[−1.519, 2.134]	[−1.891, 1.755]
α_2	3.595**	3.361**	3.642**
	(1.588)	(1.381)	(1.489)
	[0.050, 7.259]	[0.558, 6.156]	[0.586, 6.684]
β_1	−1.026***	−0.550	−0.674
	(0.371)	(0.540)	(0.491)
	[−1.889, −0.175]	[−1.512, 0.486]	[−1.757, 0.417]
β_2	−0.279	1.004	0.726
	(0.875)	(0.980)	(0.833)
	[−1.293, 0.767]	[−0.408, 2.446]	[−0.750, 2.228]
γ_1	0.133	0.038	0.000
	(0.232)	(0.179)	(0.188)
	[−0.393, 0.641]	[−0.490, 0.570]	[−0.583, 0.603]
γ_2	0.045	−0.347	−0.173
	(0.311)	(0.236)	(0.214)
	[−0.546, 0.662]	[−0.990, 0.289]	[−0.836, 0.500]
$\gamma_1 + \gamma_2$	0.178	−0.309*	−0.172
	(0.187)	(0.180)	(0.126)
	[−0.128, 0.490]	[−0.704, 0.063]	[−0.544, 0.196]
κ_1	0.206*	0.103	0.179
	(0.112)	(0.172)	(0.163)
	[−0.026, 0.446]	[−0.210, 0.427]	[−0.165, 0.523]
κ_2	−0.215	−0.473	−0.471
	(0.256)	(0.422)	(0.398)
	[−0.506, 0.067]	[−0.910, −0.050]	[−0.962, 0.009]
$\kappa_1 + \kappa_2$	−0.009	−0.370	−0.291
	(0.222)	(0.335)	(0.303)
	[−0.199, 0.162]	[−0.701, −0.051]	[−0.634, 0.059]
δ_1	0.142	0.093	0.187
	(0.187)	(0.202)	(0.178)
	[−0.107, 0.403]	[−0.187, 0.354]	[−0.095, 0.467]
δ_2	−0.192	−0.414	−0.439*
	(0.217)	(0.275)	(0.248)
	[−0.577, 0.198]	[−0.828, 0.023]	[−0.884, 0.010]
$\delta_1 + \delta_2$	−0.050	−0.322**	−0.251**
	(0.164)	(0.133)	(0.120)
	[−0.349, 0.259]	[−0.648, 0.024]	[−0.587, 0.096]
L3: Reject H_0: $\gamma_1 = \kappa_1 = \delta_1 = 0$?	F=2.732	F=0.326	F=0.642
	p=0.043	p=0.807	p=0.590
	p_{MBB}=0.089	p_{MBB}=0.869	p_{MBB}=0.745
L4: Reject H_0: $\gamma_1 + \gamma_2 = \kappa_1 + \kappa_2 = \delta_1 + \delta_2 = 0$?	F=0.322	F=2.865	F=2.359
	p=0.809	p=0.038	p=0.078
	p_{MBB}=0.861	p_{MBB}=0.214	p_{MBB}=0.390

Notes: Estimates of equation (9), which adds one-month lags of inflation, the three-month Treasury bill rate, and the unemployment rate to the regression of Table 6; γ, κ, and δ are their respective coefficients, and regimes and subscripts are as in Table 5. Prewhitened Newey–West standard errors in parentheses. Brackets: residual moving-block bootstrap (MBB) 95% percentile intervals from 4,999 replications. L3 and L4: prewhitened Newey–West F statistic and its p-value, and null-imposed residual-MBB p-value p_{MBB} from 1,999 replications. Samples as in Table 5. Stars from Newey–West t-tests: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 9: Tests of Adaptive Expectations across Regimes

	MSC	SPF	Livingston
$\beta_1(1)$	0.305^{***} (0.038) [0.227, 0.382]	0.284^{***} (0.070) [0.185, 0.388]	0.302^{***} (0.060) [0.185, 0.410]
$\beta_2(1)$	0.079^{***} (0.030) [0.033, 0.125]	-0.042 (0.040) [−0.102, 0.019]	-0.109^{***} (0.031) [−0.175, −0.040]
$\beta_1(1)+\beta_2(1)$	0.383^{***} (0.019) [0.329, 0.437]	0.242^{***} (0.040) [0.168, 0.317]	0.193^{***} (0.044) [0.112, 0.271]
α	2.059^{***} (0.150) [1.760, 2.341]	0.329 (0.325) [−0.072, 0.731]	0.278 (0.309) [−0.143, 0.670]
λ_1	-0.078^{***} (0.019) [−0.145, −0.008]	0.118 (0.156) [−0.166, 0.402]	0.013 (0.144) [−0.373, 0.392]
λ_2	0.115^{***} (0.023) [0.045, 0.182]	0.039 (0.138) [−0.225, 0.304]	0.158 (0.150) [−0.203, 0.525]
θ_1	−0.046 (0.117) [−0.202, 0.116]	0.321^{**} (0.143) [0.046, 0.584]	0.727^{**} (0.298) [0.192, 1.258]
θ_2	0.009 (0.111) [−0.153, 0.166]	−0.082 (0.136) [−0.347, 0.195]	-0.494^{*} (0.286) [−1.006, 0.029]
L5: Reject H_0: $\lambda_1=\lambda_2=\theta_1=\theta_2=0$?	F=9.261 p<0.001 p_{MBB}=0.020	F=31.026 p<0.001 p_{MBB}=0.001	F=22.023 p<0.001 p_{MBB}=0.002

Notes: Estimates of equation (10), a regression of the survey forecast on lagged inflation and controls, with regimes as in Table 5. Prewhitened Newey–West standard errors in parentheses. $\beta_1(1)$ is the stable-regime cumulative loading, the sum of the twelve monthly inflation-lag coefficients. $\beta_2(1)$ is its volatile-regime shift, and $\beta_1(1)+\beta_2(1)$ is the volatile-regime cumulative loading. α is the intercept, which does not vary by regime. The controls λ_1 and λ_2 multiply current and lagged unemployment, θ_1 and θ_2 current and lagged three-month Treasury bill rates. Brackets: residual moving-block bootstrap (MBB) 95% percentile intervals from 4,999 replications. L5: prewhitened Newey–West F statistic and its p-value, and null-imposed residual-MBB p-value p_{MBB} from 1,999 replications. Samples: MSC 520 monthly (1981:M9–2024:M12), SPF 174 quarterly, Livingston 87 semiannual. Stars from Newey–West t-tests: $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

in the rationality regressions of Section 5.1.[13]

Third, none of the surveys is purely backward-looking. The macroeconomic controls are jointly significant for all three surveys, with unemployment terms individually significant for households and interest-rate terms for the professionals.[14]

[13]With a regime-specific intercept, the household shift keeps its size, at 0.076, but loses significance, because the slope and intercept shifts are nearly collinear. The two household shifts are jointly significant at the 5% level under Newey–West but only at the 10% level under the null-imposed bootstrap, so we read the household evidence in this specification as suggestive. The professional slope shifts turn positive, at 0.128 for the SPF and 0.048 for Livingston.

[14]The controls remain jointly significant when we re-estimate the regression without regime terms, as Table D.2 of Appendix D reports.

5.3 Discussion

When inflation is stable, all three surveys put significant weight on recently realized inflation, and none of them shows significant bias or own-forecast inefficiency. When inflation turns volatile, households raise the weight they place on recently realized inflation, whereas neither professional survey shows a robust change in that weight. This contrast is the paper's central finding. Household expectations therefore respond more strongly to recent inflation exactly when inflation moves the most, which is also when expectations matter most for wage bargaining, durable goods purchases, and monetary policy.

The departures from rationality also depend on the regime. Those specific to households, persistent errors and unexploited macroeconomic information, occur in the stable regime and are only suggestive. In the volatile regime, own-forecast inefficiency is robust for the professionals and only suggestive for households. Conditioning on the regime makes this asymmetry visible.[15] The departures therefore do not map onto a simple divide between uninformed households and informed professionals.

Read together with Section 4, these results separate forecast accuracy from expectation formation. Neither group can be shown to forecast more accurately, yet households raise the weight they place on recently realized inflation when inflation turns volatile. For central banks, accuracy therefore gives no reason to set household surveys aside in favor of professional forecasts. The regime-dependent household weight further supports the joint monitoring of household and professional forecasts, particularly in volatile periods. For applied work, a Phillips curve or a monetary-policy rule estimated with household expectations inherits a weight on recent inflation that depends on the regime, so the choice of expectation measure is a modeling assumption to make explicit.

6 Robustness Checks

This section checks four results that our conclusions rest on: the loss-difference tests under squared loss, the professional own-forecast efficiency in both regimes, the cumulative loadings $\beta_1(1)$ and $\beta_1(1) + \beta_2(1)$, and the regime shifts $\beta_2(1)$. The first three survive every check that bears on them. The household shift stays positive under every check of this section but loses significance under two of them, the semiannual Livingston dates and the lagged filtered indicator. The SPF shift is insignificant under

[15] Without regime interactions, the tests reproduce familiar full-sample results, reported in Table D.1 of Appendix D. The own-forecast efficiency test rejects for both professional surveys but not for the MSC, and the macroeconomic-information test rejects for no survey.

Table 10: Common-Date Robustness of the Household Loadings

Result	MSC at SPF dates (quarterly)	MSC at Livingston dates (semiannual)
Cumulative loading, stable ($\beta_1(1)$)	0.303 ($p < 0.001$)	0.295 ($p < 0.001$)
Cumulative loading, volatile ($\beta_1(1) + \beta_2(1)$)	0.368 ($p < 0.001$)	0.342 ($p < 0.001$)
Cumulative loading, regime shift ($\beta_2(1)$)	0.065 ($p = 0.041$)	0.047 ($p = 0.190$)

Notes: The frequency check of Section 6.1: the MSC cumulative loadings of the Table 9 specification and their regime shift, re-estimated on the professional surveys' observation dates, 174 quarterly and 87 semiannual, with the Newey–West bandwidth reset to one year at each frequency, four and two lags. Cells report the estimate with its prewhitened Newey–West p-value in parentheses.

every regime classification, and the Livingston shift, although significant under every classification by the Newey–West route, has an interval under the re-estimation of the regimes that includes zero.

Section 6.1 re-estimates the household loadings on the professional surveys' observation dates. Section 6.2 re-runs the core results under six regime classifications. Section 6.3 compares the prewhitened estimator with its alternatives, and Section 6.4 propagates the first-stage uncertainty of the estimated regime into the results. Appendix E collects the supporting tables and the bootstrap procedures.

6.1 Frequency and common observation dates

Two of the four target results need no frequency check. We compute the loss differences on the common observation dates by construction, so Table 3 is already the common-date result, and we estimate the professional efficiency and loading results at the professionals' own dates throughout. The check therefore bears on the household side. Table 10 re-estimates the MSC cumulative loadings and their regime shift on the professional surveys' observation dates, 174 quarterly and 87 semiannual, with the HAC bandwidth reset to one year at each frequency.

The loadings survive. The stable-regime loading $\beta_1(1)$ is 0.303 quarterly and 0.295 semiannual, the volatile-regime sum $\beta_1(1) + \beta_2(1)$ is 0.368 and 0.342, and all four are significant at the 1% level. The regime shift survives at quarterly frequency but not at semiannual frequency. On the quarterly SPF dates, $\beta_2(1)$ is 0.065 and significant at the 5% level. On the semiannual Livingston dates, it is 0.047 with a p-value of 0.19. With 87 observations, only 32 of them in the volatile regime, the semiannual sample cannot establish the shift by itself. Its point estimate is also smaller than the monthly 0.079.

6.2 Alternative regime definitions

The baseline indicator of equation (3) applies a cutoff of one-half to smoothed probabilities, which condition on the full sample, so it is an *ex post* description of the inflation environment. An *ex post* description is the right object for our question, which asks how expectations behaved in environments that were in fact turbulent, but the indicator embeds choices, and the smoothed probability at t draws on inflation realized after t. We therefore consider six classifications, the baseline plus five variants. The first replaces the binary indicator with the continuous smoothed probability. The second uses filtered rather than smoothed probabilities, which condition only on information up to t and so remove the look-ahead. The third and fourth move the cutoff to $c = 0.4$ and $c = 0.6$ around the baseline $c = 0.5$, and the fifth dates the filtered indicator one quarter before the forecast origin. The filtered probability at t still conditions on inflation realized during the origin quarter, which the expectations formed at t both observe and can feed into. Dating the filtered indicator one quarter earlier also removes this contemporaneous feedback, and the resulting variant is deliberately the noisiest classification. The cutoff variants, by contrast, can differ only at the margin. Just seven of the 174 evaluation-window quarters carry a smoothed probability between 0.4 and 0.6, so the three cutoffs draw nearly the same line.

Tables 11 and 12 report each core quantity under each classification. The professional efficiency contrast is the most robust result. The volatile-regime sum is significant at the 5% level or better under every classification for both surveys, including the lagged filtered indicator, where the sums are -0.705 for the SPF and -0.702 for Livingston. The stable-regime coefficient β_1 stays insignificant under every classification, with the smallest p-value at 0.33, so the inefficiency appears in the volatile regime and only there.

The volatile-regime cumulative loadings barely move across classifications. The household loading stays between 0.371 and 0.383, the SPF loading between 0.232 and 0.246, and the Livingston loading between 0.171 and 0.196, every entry significant at the 1% level with a residual-MBB interval strictly above zero. The stable-regime loadings stay between 0.250 and 0.334 across surveys and classifications, each significant at the 1% level.

The regime shifts $\beta_2(1)$ keep their baseline signs under all six classifications. The household shift is significant under five of them, with a residual-MBB interval above zero in each of the five. The exception is the lagged filtered indicator, where the shift falls to an insignificant 0.037 with a standard error of 0.030. The same column shows why. Under this variant, the volatile-regime loss differences collapse toward zero, to -0.008 and 0.198 from -0.592 and -0.588 at the baseline, so the lagged filtered clas-

sification no longer separates turbulent from tranquil environments. Attenuation toward zero is what misclassification of a binary conditioning variable produces, and the loading levels hold steady through it. The SPF shift stays negative and insignificant throughout. The Livingston shift is negative and significant under all six by the Newey–West route, but under the continuous classification its residual-MBB interval, $[-0.196,\ 0.007]$, reaches just above zero.

On the accuracy side, no mean loss difference is significant under any classification, in either regime. Signs match the baseline everywhere except in the MSC–Livingston rows of the lagged filtered column, where the volatile-regime difference turns slightly positive and the stable-regime difference slightly negative.

The filtered-probability column is the closest any variant comes to real-time construction with contemporaneous data. Under the filtered indicator, the professional volatile-regime sums are -0.703 for the SPF and -0.721 for Livingston, both significant at the 1% level, and the household loading shift is 0.066, significant at the 5% level. An operational reading can therefore rest on the professional inefficiency under both filtered variants, and on the household loading shift under the unlagged filtered variant only.

6.3 HAC estimator and block-bootstrap inference

The serial correlation that the twelve-month overlap builds into the errors is strong, with a first-order autocorrelation of 0.74 in the quarterly loss difference, and kernel-only HAC estimators over-reject when correlation is this strong. The baseline of Section 2.3 therefore prewhitens, applying a first-order vector autoregressive filter before the Bartlett kernel. Prewhitening removes most of the size distortion, and it is also the conservative choice here, raising the standard error relative to the non-prewhitened estimator in nearly every series we examine. This subsection checks the baseline four ways.

Serial correlation. Table E.1 in Appendix E reports the loss-difference autocorrelations. Those up to lag q, the moving-average order that the twelve-month overlap implies, are significant in both squared-loss series, as the overlap predicts. Beyond lag q, a joint test on lags $q+1$ through $q+4$ fails to reject in either series. The fixed one-year bandwidth and block length are therefore adequate.

Prewhitening. We recompute every main-table standard error and joint test with the non-prewhitened Newey–West estimator, point estimates fixed. On the four targets, the loss differences stay insignificant. The professional efficiency contrast of Table 6 is intact, with smaller standard errors on the volatile-regime sums and with the stable-

Table 11: Regime-Definition Robustness: Accuracy

Result	Continuous probability	Filtered probability	Filtered, lagged one quarter
MSC−SPF loss difference, stable	0.602 (0.594) [−0.151, 1.351]	0.150 (0.446) [−0.541, 0.792]	0.098 (0.522) [−0.806, 0.848]
MSC−SPF loss difference, volatile	−0.904 (2.818) [−4.122, 2.656]	−0.190 (2.181) [−2.752, 2.815]	−0.008 (1.567) [−1.960, 2.801]
MSC−Livingston loss difference, stable	0.510 (0.448) [−0.212, 1.236]	0.134 (0.353) [−0.406, 0.675]	−0.085 (0.588) [−1.223, 0.773]
MSC−Livingston loss difference, volatile	−0.843 (2.033) [−4.444, 2.983]	−0.252 (1.749) [−3.408, 3.065]	0.198 (1.050) [−1.632, 2.866]
Result	$c = 0.4$	$c = 0.5$ (baseline)	$c = 0.6$
MSC−SPF loss difference, stable	0.494 (0.380) [−0.049, 1.072]	0.405 (0.410) [−0.173, 0.993]	0.345 (0.456) [−0.266, 0.958]
MSC−SPF loss difference, volatile	−0.655 (2.063) [−3.027, 2.020]	−0.592 (2.185) [−3.120, 2.233]	−0.516 (2.174) [−3.113, 2.334]
MSC−Livingston loss difference, stable	0.426 (0.330) [−0.020, 0.929]	0.357 (0.321) [−0.131, 0.885]	0.268 (0.381) [−0.253, 0.823]
MSC−Livingston loss difference, volatile	−0.641 (1.498) [−3.241, 2.208]	−0.588 (1.538) [−3.327, 2.397]	−0.481 (1.559) [−3.430, 2.638]

Notes: The regime-definition check of Section 6.2, accuracy side: mean loss differences $\bar{d}$ under squared loss by regime, negative entries indicating lower household loss; no entry is significantly different from zero at the 10% level. Each cell reports the estimate, its prewhitened Newey–West standard error in parentheses, and a moving-block bootstrap 95% percentile interval in brackets from 9,999 replications, resampling the loss difference and the classification jointly. Columns headed c use the cutoff c on smoothed volatile-regime probabilities; the baseline column ($c = 0.5$) reproduces the squared-loss rows of Table 3; the filtered-probability column uses the 0.5 cutoff on filtered probabilities; the lagged filtered column dates that indicator one quarter before the forecast origin, which drops the first quarterly common date and leaves 173 MSC–SPF and 87 MSC–Livingston observations; under the continuous specification we replace the regime dummy with the smoothed probability p_t and evaluate the volatile-regime mean at $p_t = 1$ and the stable-regime mean at $p_t = 0$.

Table 12: Regime-Definition Robustness: Rationality and Formation

Result	Continuous probability	Filtered probability	Filtered, lagged one quarter
SPF own-forecast efficiency, stable (β_1)	-0.153	-0.325	-0.350
	(0.334)	(0.380)	(0.415)
	$[-0.712, 0.457]$	$[-0.794, 0.226]$	$[-0.829, 0.180]$
Livingston own-forecast efficiency, stable (β_1)	-0.116	-0.291	-0.338
	(0.284)	(0.347)	(0.377)
	$[-0.715, 0.522]$	$[-0.802, 0.282]$	$[-0.827, 0.210]$
SPF own-forecast efficiency, volatile ($\beta_1 + \beta_2$)	-0.890^{**}	-0.703^{***}	-0.705^{***}
	(0.352)	(0.230)	(0.193)
	$[-1.324, -0.443]$	$[-1.098, -0.244]$	$[-1.138, -0.215]$
Livingston own-forecast efficiency, volatile ($\beta_1 + \beta_2$)	-0.924^{***}	-0.721^{***}	-0.702^{***}
	(0.284)	(0.198)	(0.117)
	$[-1.393, -0.361]$	$[-1.138, -0.208]$	$[-1.134, -0.157]$
MSC cumulative loading, stable ($\beta_1(1)$)	0.250^{***}	0.317^{***}	0.334^{***}
	(0.043)	(0.045)	(0.043)
	[0.156, 0.345]	[0.229, 0.403]	[0.246, 0.419]
SPF cumulative loading, stable ($\beta_1(1)$)	0.285^{***}	0.288^{***}	0.288^{***}
	(0.094)	(0.073)	(0.073)
	[0.160, 0.414]	[0.177, 0.399]	[0.176, 0.394]
Livingston cumulative loading, stable ($\beta_1(1)$)	0.265^{***}	0.262^{***}	0.271^{***}
	(0.079)	(0.064)	(0.073)
	[0.117, 0.406]	[0.144, 0.375]	[0.154, 0.384]
MSC cumulative loading, volatile ($\beta_1(1) + \beta_2(1)$)	0.373^{***}	0.383^{***}	0.371^{***}
	(0.019)	(0.023)	(0.025)
	[0.317, 0.428]	[0.325, 0.440]	[0.311, 0.430]
SPF cumulative loading, volatile ($\beta_1(1) + \beta_2(1)$)	0.242^{***}	0.244^{***}	0.232^{***}
	(0.046)	(0.043)	(0.042)
	[0.167, 0.318]	[0.168, 0.320]	[0.158, 0.310]
Livingston cumulative loading, volatile ($\beta_1(1) + \beta_2(1)$)	0.171^{***}	0.182^{***}	0.189^{***}
	(0.049)	(0.044)	(0.047)
	[0.089, 0.252]	[0.103, 0.261]	[0.107, 0.269]
MSC cumulative loading, regime shift ($\beta_2(1)$)	0.123^{***}	0.066^{**}	0.037
	(0.034)	(0.030)	(0.030)
	[0.061, 0.187]	[0.017, 0.116]	$[-0.010, 0.083]$
SPF cumulative loading, regime shift ($\beta_2(1)$)	-0.043	-0.044	-0.056
	(0.059)	(0.039)	(0.038)
	$[-0.130, 0.044]$	$[-0.108, 0.018]$	$[-0.118, 0.008]$
Livingston cumulative loading, regime shift ($\beta_2(1)$)	-0.094^{**}	-0.080^{***}	-0.082^{**}
	(0.046)	(0.030)	(0.038)
	$[-0.196, 0.007]$	$[-0.146, -0.011]$	$[-0.149, -0.015]$

Notes: The table continues on the next page with the cutoff classifications, where the full notes appear. Stars from Newey–West t-tests: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 12: Regime-Definition Robustness: Rationality and Formation (continued)

Result	$c = 0.4$	$c = 0.5$ (baseline)	$c = 0.6$
SPF own-forecast efficiency, stable (β_1)	-0.185	-0.230	-0.299
	(0.232)	(0.272)	(0.330)
	$[-0.624, 0.310]$	$[-0.681, 0.274]$	$[-0.750, 0.222]$
Livingston own-forecast efficiency, stable (β_1)	-0.185	-0.173	-0.292
	(0.212)	(0.222)	(0.296)
	$[-0.664, 0.320]$	$[-0.641, 0.335]$	$[-0.757, 0.235]$
SPF own-forecast efficiency, volatile ($\beta_1 + \beta_2$)	-0.846^{***}	-0.831^{***}	-0.787^{***}
	(0.324)	(0.288)	(0.251)
	$[-1.223, -0.460]$	$[-1.223, -0.415]$	$[-1.178, -0.355]$
Livingston own-forecast efficiency, volatile ($\beta_1 + \beta_2$)	-0.868^{***}	-0.871^{***}	-0.781^{***}
	(0.278)	(0.245)	(0.188)
	$[-1.267, -0.423]$	$[-1.285, -0.399]$	$[-1.211, -0.258]$
MSC cumulative loading, stable ($\beta_1(1)$)	0.309^{***}	0.305^{***}	0.303^{***}
	(0.040)	(0.038)	(0.039)
	$[0.228, 0.390]$	$[0.227, 0.382]$	$[0.224, 0.380]$
SPF cumulative loading, stable ($\beta_1(1)$)	0.301^{***}	0.284^{***}	0.284^{***}
	(0.071)	(0.070)	(0.071)
	$[0.197, 0.406]$	$[0.185, 0.388]$	$[0.185, 0.388]$
Livingston cumulative loading, stable ($\beta_1(1)$)	0.293^{***}	0.302^{***}	0.304^{***}
	(0.060)	(0.060)	(0.066)
	$[0.173, 0.410]$	$[0.185, 0.410]$	$[0.188, 0.415]$
MSC cumulative loading, volatile ($\beta_1(1) + \beta_2(1)$)	0.383^{***}	0.383^{***}	0.382^{***}
	(0.020)	(0.019)	(0.019)
	$[0.329, 0.437]$	$[0.329, 0.437]$	$[0.328, 0.436]$
SPF cumulative loading, volatile ($\beta_1(1) + \beta_2(1)$)	0.246^{***}	0.242^{***}	0.242^{***}
	(0.042)	(0.040)	(0.040)
	$[0.173, 0.320]$	$[0.168, 0.317]$	$[0.169, 0.317]$
Livingston cumulative loading, volatile ($\beta_1(1) + \beta_2(1)$)	0.187^{***}	0.193^{***}	0.196^{***}
	(0.044)	(0.044)	(0.045)
	$[0.107, 0.267]$	$[0.112, 0.271]$	$[0.115, 0.275]$
MSC cumulative loading, regime shift ($\beta_2(1)$)	0.074^{**}	0.079^{***}	0.079^{***}
	(0.029)	(0.030)	(0.030)
	$[0.027, 0.122]$	$[0.033, 0.125]$	$[0.035, 0.125]$
SPF cumulative loading, regime shift ($\beta_2(1)$)	-0.055	-0.042	-0.041
	(0.038)	(0.040)	(0.040)
	$[-0.118, 0.007]$	$[-0.102, 0.019]$	$[-0.102, 0.017]$
Livingston cumulative loading, regime shift ($\beta_2(1)$)	-0.106^{***}	-0.109^{***}	-0.108^{***}
	(0.033)	(0.031)	(0.033)
	$[-0.176, -0.034]$	$[-0.175, -0.040]$	$[-0.175, -0.041]$

Notes: The regime-definition check of Section 6.2, formation side: own-forecast efficiency rows report the stable-regime coefficient β_1 and the volatile-regime sum $\beta_1 + \beta_2$, cumulative-loading rows the stable-regime loading $\beta_1(1)$, the volatile-regime loading $\beta_1(1) + \beta_2(1)$, and the regime shift $\beta_2(1)$. Each cell reports the estimate, its prewhitened Newey–West standard error in parentheses, and a residual moving-block bootstrap 95% percentile interval in brackets from 4,999 replications. Columns headed c use the cutoff c on smoothed volatile-regime probabilities; the baseline column ($c = 0.5$) reproduces Tables 6 and 9; the filtered-probability column uses the 0.5 cutoff on filtered probabilities; the lagged filtered column dates that indicator one quarter before the forecast origin, so the first three window months drop out and the cumulative-loading rows rest on 517 MSC, 173 SPF, and 87 Livingston observations; under the continuous specification we replace the regime dummy with the smoothed probability p_t, evaluating volatile-regime quantities at $p_t = 1$ and stable-regime quantities at $p_t = 0$, so the regime shift is the change from $p_t = 0$ to $p_t = 1$. Stars from Newey–West t-tests: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

regime coefficients still insignificant. All of Table 9, including the household shift, survives.

Small-sample corrections. The baseline takes p-values from a t distribution but applies neither the $n/(n-k)$ covariance scaling nor the small-sample correction of Harvey et al. (1997), hereafter HLN, to the Diebold–Mariano statistic. Table E.2 adds both to every squared-loss cell, and neither changes a conclusion. The HLN correction lowers the p-value in most cells and raises it in one by 0.018, so the paper's convention is generally the more conservative. The smallest p-value under any combination is 0.224.

Moving-block bootstrap. The bracketed intervals and p_{MBB} entries throughout our tables come from a moving-block bootstrap, a second inference route that bypasses HAC asymptotics. Section 2.3 introduces it, and Appendix E.3 gives the mechanics. Table E.3 collects the headline quantities, and the bootstrap confirms them. Its intervals exclude zero for the professional volatile-regime inefficiency, for the stable and volatile cumulative loadings of all three surveys, and for the household loading shift. They straddle zero for the stable-regime efficiency coefficients, in line with Table 6.

6.4 Regime-estimation uncertainty

The regime indicator is a generated regressor. We estimate it in a first stage, and second-stage standard errors that treat it as known omit the first-stage uncertainty. We quantify the omission with a parametric bootstrap of the Markov-switching model, parametric because each replication simulates a fresh inflation history from the estimated MS-AR(1) rather than resampling blocks of data. Each replication re-estimates the model on its simulated history, uses the re-estimated parameters to reclassify the actual series at the baseline cutoff, and re-runs the second stage. Holding the data fixed and varying only the classification isolates the one channel through which first-stage uncertainty reaches the results. If the regime parameters were poorly pinned down, the reclassified indicators would disagree across replications and the second-stage estimates would spread. Table 13 reports 999 replications, all successful. Appendix E.4 details the design.

Each interval in Table 13 measures how far the estimated classification can move an estimate, so an interval that excludes zero signals a stable estimate, not a significant one. The classification is stable. The median replication retains the baseline label for 93.1% of evaluation-window months. The professional volatile-regime sums are essentially unaffected, with intervals $[-0.897, -0.573]$ and $[-0.930, -0.524]$ strictly below zero. The stable-regime coefficients stay negative and moderate as the classification varies, and the volatile-regime sum lies below the stable coefficient in 97.6% of

Table 13: Parametric Markov-Switching Bootstrap: Core Quantities

Result	Baseline estimate	Bootstrap 95% interval
MSC−SPF loss difference, stable	0.405	[0.073, 0.630]
MSC−SPF loss difference, volatile	−0.592	[−1.185, 0.031]
MSC−Livingston loss difference, stable	0.357	[0.018, 0.577]
MSC−Livingston loss difference, volatile	−0.588	[−1.037, 0.009]
SPF own-forecast efficiency, stable (β_1)	−0.230	[−0.663, −0.088]
Livingston own-forecast efficiency, stable (β_1)	−0.173	[−0.618, −0.036]
SPF own-forecast efficiency, volatile ($\beta_1 + \beta_2$)	−0.831	[−0.897, −0.573]
Livingston own-forecast efficiency, volatile ($\beta_1 + \beta_2$)	−0.871	[−0.930, −0.524]
MSC cumulative loading, stable ($\beta_1(1)$)	0.305	[0.258, 0.389]
SPF cumulative loading, stable ($\beta_1(1)$)	0.284	[0.176, 0.315]
Livingston cumulative loading, stable ($\beta_1(1)$)	0.302	[0.135, 0.309]
MSC cumulative loading, volatile ($\beta_1(1) + \beta_2(1)$)	0.383	[0.371, 0.432]
SPF cumulative loading, volatile ($\beta_1(1) + \beta_2(1)$)	0.242	[0.198, 0.254]
Livingston cumulative loading, volatile ($\beta_1(1) + \beta_2(1)$)	0.193	[0.118, 0.206]
MSC cumulative loading, regime shift ($\beta_2(1)$)	0.079	[0.027, 0.121]
SPF cumulative loading, regime shift ($\beta_2(1)$)	−0.042	[−0.068, 0.045]
Livingston cumulative loading, regime shift ($\beta_2(1)$)	−0.109	[−0.117, 0.021]

Notes: The regime-uncertainty check of Section 6.4: baseline estimates from Tables 3, 6, and 9, with the loss differences under squared loss. Each interval reports the 2.5–97.5 percentiles across 999 parametric-bootstrap replications of the first-stage MS-AR(1), each entry using the 978–999 replications for which the quantity is defined; the data are held fixed and only the regime classification varies, following the four-step procedure of Appendix E.4, so the intervals reflect first-stage regime-estimation uncertainty, not sampling uncertainty. The median replication retains the baseline classification for 93.1% of evaluation-window months.

replications for the SPF and 97.0% for Livingston, so the gap between the two regimes survives the first-stage uncertainty. The cumulative loadings are just as stable, with volatile-regime intervals of [0.371, 0.432] for the MSC, [0.198, 0.254] for the SPF, and [0.118, 0.206] for Livingston. The household regime shift is equally stable, with an interval of [0.027, 0.121] above zero, whereas both professional intervals include zero, at [−0.068, 0.045] for the SPF and [−0.117, 0.021] for Livingston. The stable-regime loss differences have intervals above zero, yet Table 3 shows them insignificant. The volatile-regime loss differences remain negative in 97% of replications, with upper endpoints just above zero. Sampling uncertainty, not first-stage uncertainty, keeps the accuracy reversal from significance.

7 Conclusion

This paper asks whether professional inflation forecasts are more accurate than household expectations, and whether the way each group forms expectations changes with the inflation environment. We compare median one-year-ahead expectations from the MSC, the SPF, and the Livingston Survey over 1981–2024, separating stable from volatile environments with a two-state Markov-switching model of realized CPI inflation.

The two questions have different answers. On accuracy, neither group can be shown to outperform the other. On expectation formation, the household weight on recent inflation and the departures from rationality both depend on the regime. When inflation becomes volatile, households put more weight on recently realized inflation, while neither professional survey shows a robust change in that weight. The rise in the household weight is robust to alternative standard errors, to most alternative regime definitions, to uncertainty in the estimated regime, and to ending the sample before the pandemic. Volatile-regime errors also become predictable from the forecasts themselves, a result that is robust for the professionals and only suggestive for households. The accuracy results and the professional inefficiency survive every check that bears on them.

Accuracy therefore cannot settle the choice of expectation measure. For central banks, accuracy gives no reason to set household surveys aside, and the household weight on recent inflation depends on the regime. Both findings support the joint monitoring of household and professional forecasts, particularly in volatile periods. In applied work, which measure enters a Phillips curve or a monetary-policy rule is a modeling assumption to state and justify.

Several limitations point to future work. Aggregate survey medians cannot speak to individual updating, disagreement, or compositional change. The estimated regime is a conditioning variable, not a treatment, so the results are descriptive rather than causal. The evidence, moreover, covers one country, one horizon, and few volatile-regime observations. Individual-level expectations, other countries and horizons, and direct measures of information acquisition could identify the mechanisms behind the state dependence documented here.

References

Andrews DWK, Monahan JC (1992) An improved heteroskedasticity and autocorrelation consistent covariance matrix estimator. Econometrica 60:953-966. https://doi.org/10.2307/2951574

Ang A, Bekaert G, Wei M (2007) Do macro variables, asset markets, or surveys forecast inflation better? Journal of Monetary Economics 54:1163-1212. https://doi.org/10.1016/j.jmoneco.2006.04.006

Benchimol J, El-Shagi M, Saadon Y (2022) Do expert experience and characteristics affect inflation forecasts? Journal of Economic Behavior & Organization 201:205-226. https://doi.org/10.1016/j.jebo.2022.06.025

Bernanke BS (2007) Inflation expectations and inflation forecasting. Speech at the Monetary Economics Workshop of the NBER Summer Institute, Cambridge, MA, July 10. https://www.federalreserve.gov/newsevents/speech/bernanke20070710a.htm

Bollerslev T (1986) Generalized autoregressive conditional heteroskedasticity. Journal of Econometrics 31:307-327. https://doi.org/10.1016/0304-4076(86)90063-1

Bordalo P, Gennaioli N, Ma Y, Shleifer A (2020) Overreaction in macroeconomic expectations. American Economic Review 110:2748-2782. https://doi.org/10.1257/aer.20181219

Bracha A, Tang J (2025) Inflation levels and (in)attention. The Review of Economic Studies 92:1564-1594. https://doi.org/10.1093/restud/rdae063

Burke MA, Ozdagli A (2023) Household inflation expectations and consumer spending: evidence from panel data. The Review of Economics and Statistics 105:948-961. https://doi.org/10.1162/rest_a_01118

Candelon B, Roccazzella F (2025) Evaluating inflation forecasts in the euro area and the role of the ECB. Journal of Forecasting 44:978-1008. https://doi.org/10.1002/for.3235

Coibion O, Gorodnichenko Y (2015) Information rigidity and the expectations formation process: a simple framework and new facts. American Economic Review 105:2644-2678. https://doi.org/10.1257/aer.20110306

D'Acunto F, Malmendier U, Ospina J, Weber M (2021) Exposure to grocery prices and inflation expectations. Journal of Political Economy 129:1615-1639. https://doi.org/10.1086/713192

D'Acunto F, Malmendier U, Weber M (2023) What do the data tell us about inflation expectations? In: Bachmann R, Topa G, van der Klaauw W (eds) Handbook of economic expectations. Elsevier, Amsterdam, pp 133-161. https://doi.org/10.1016/B978-0-12-822927-9.00012-4

Diebold FX, Mariano RS (1995) Comparing predictive accuracy. Journal of Business & Economic Statistics 13:253-263. https://doi.org/10.1080/07350015.1995.10524599

El-Shagi M (2011) Inflation expectations: does the market beat econometric forecasts? The North American Journal of Economics and Finance 22:298-319. https://doi.org/10.1016/j.najef.2011.05.002

Evans M, Wachtel P (1993) Inflation regimes and the sources of inflation uncertainty. Journal of Money, Credit and Banking 25:475-511. https://doi.org/10.2307/2077719

Giacomini R, Rossi B (2010) Forecast comparisons in unstable environments. Journal of Applied Econometrics 25:595-620. https://doi.org/10.1002/jae.1177

Giacomini R, White H (2006) Tests of conditional predictive ability. Econometrica 74:1545-1578. https://doi.org/10.1111/j.1468-0262.2006.00718.x

Goodspeed T (2025) Trust the experts? The performance of inflation expectations, 1960–2023. International Journal of Forecasting 41:863-876. https://doi.org/10.1016/j.ijforecast.2024.06.006

Gwak B (2022) State-dependent formation of inflation expectations. Macroeconomic Dynamics 26:1994-2028. https://doi.org/10.1017/S1365100521000171

Hamilton JD (1989) A new approach to the economic analysis of nonstationary time series and the business cycle. Econometrica 57:357-384. https://doi.org/10.2307/1912559

Hansen BE (1992) Testing for parameter instability in linear models. Journal of Policy Modeling 14:517-533. https://doi.org/10.1016/0161-8938(92)90019-9

Harvey D, Leybourne S, Newbold P (1997) Testing the equality of prediction mean squared errors. International Journal of Forecasting 13:281-291. https://doi.org/10.1016/S0169-2070(96)00719-4

Henry E, Mulloy C, Sarte P-D (2023) What survey measures of inflation expectations tell us. Economic Brief 23-03, Federal Reserve Bank of Richmond. https://www.richmondfed.org/publications/research/economic_brief/2023/eb_23-03

Inclán C, Tiao GC (1994) Use of cumulative sums of squares for retrospective detection of changes of variance. Journal of the American Statistical Association 89:913-923. https://doi.org/10.1080/01621459.1994.10476824

Korenok O, Munro D, Chen J (2026) Inflation and attention thresholds. The Review of Economics and Statistics 108:842-850. https://doi.org/10.1162/rest_a_01402

Lamoureux CG, Lastrapes WD (1990) Persistence in variance, structural change, and the GARCH model. Journal of Business & Economic Statistics 8:225-234. https://doi.org/10.1080/07350015.1990.10509794

Lieb L, Schuffels J (2022) Inflation expectations and consumer spending: the role of household balance sheets. Empirical Economics 63:2479-2512. https://doi.org/10.1007/s00181-022-02222-8

Malmendier U, Nagel S (2016) Learning from inflation experiences. The Quarterly Journal of Economics 131:53-87. https://doi.org/10.1093/qje/qjv037

Mankiw NG, Reis R (2002) Sticky information versus sticky prices: a proposal to replace the New Keynesian Phillips curve. The Quarterly Journal of Economics 117:1295-1328. https://doi.org/10.1162/003355302320935034

Mankiw NG, Reis R, Wolfers J (2003) Disagreement about inflation expectations. NBER Macroeconomics Annual 18:209-248. https://doi.org/10.1086/ma.18.3585256

Mehra YP (2002) Survey measures of expected inflation: revisiting the issues of predictive content and rationality. Federal Reserve Bank of Richmond Economic Quarterly 88:17-36. https://www.richmondfed.org/publications/research/economic_quarterly/2002/summer/mehra

Newey WK, West KD (1987) A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. Econometrica 55:703-708. https://doi.org/10.2307/1913610

Newey WK, West KD (1994) Automatic lag selection in covariance matrix estimation. The Review of Economic Studies 61:631-653. https://doi.org/10.2307/2297912

Nyblom J (1989) Testing for the constancy of parameters over time. Journal of the American Statistical Association 84:223-230. https://doi.org/10.1080/01621459.1989.10478759

Pfajfar D, Santoro E (2010) Heterogeneity, learning and information stickiness in inflation expectations. Journal of Economic Behavior & Organization 75:426-444. https://doi.org/10.1016/j.jebo.2010.05.012

Pfäuti O (2025) Inflation—who cares? Monetary policy in times of low attention. Journal of Money, Credit and Banking 57:1211-1239. https://doi.org/10.1111/jmcb.13145

Reis R (2021) Losing the inflation anchor. Brookings Papers on Economic Activity 52:307-361. https://doi.org/10.1353/eca.2022.0004

Sansó A, Aragó V, Carrion-i-Silvestre JL (2004) Testing for changes in the unconditional variance of financial time series. Revista de Economía Financiera 4:32-53.

Sims CA (2003) Implications of rational inattention. Journal of Monetary Economics 50:665-690. https://doi.org/10.1016/S0304-3932(03)00029-1

Weber M, D'Acunto F, Gorodnichenko Y, Coibion O (2022) The subjective inflation expectations of households and firms: measurement, determinants, and implications. Journal of Economic Perspectives 36:157-184. https://doi.org/10.1257/jep.36.3.157

Weber M, Candia B, Afrouzi H, Ropele T, Lluberas R, Frache S, Meyer B, Kumar S, Gorodnichenko Y, Georgarakos D, Coibion O, Kenny G, Ponce J (2025) Tell me something I don't already know: learning in low- and high-inflation settings. Econometrica 93:229-264. https://doi.org/10.3982/ECTA22764

Appendices

A Additional Data Details

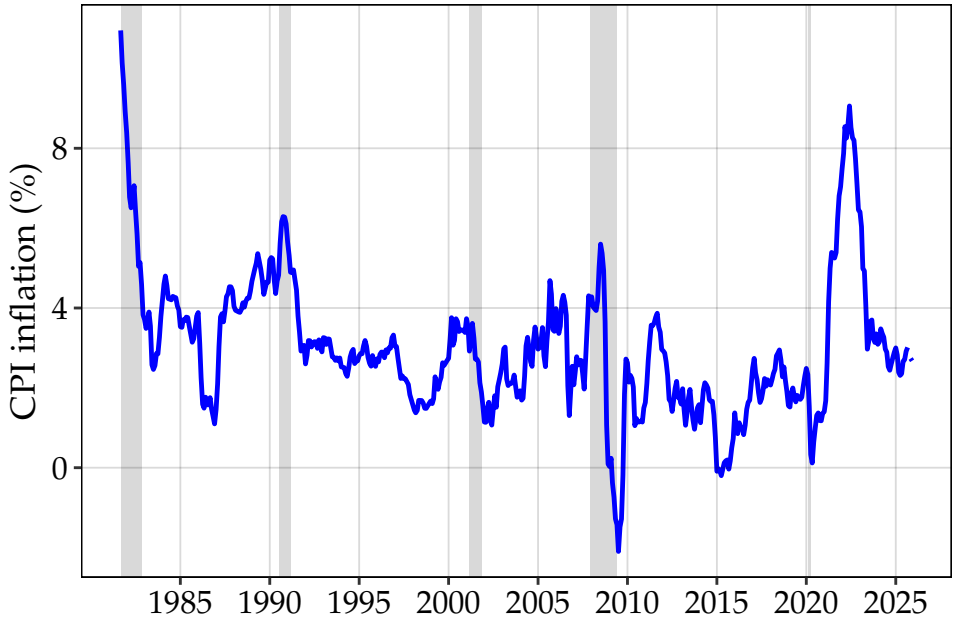
Table A.1 lists every series used in the paper, its source, and its treatment. Figure A.1 plots realized year-over-year CPI inflation over the sample period. The mean and, even more, the variability of the series shift over time, with turbulence concentrated in the early and mid-1980s, around the 1990–1991 and 2008–2009 recessions, and during the post-pandemic surge. We fit the Markov-switching model of Section 2.2 to quarterly final-month year-over-year CPI inflation over 1981:Q2–2025:Q4. All series are final-vintage values as archived in the replication package. The Treasury bill rate and the not-seasonally-adjusted CPI are not revised after release; the unemployment rate is subject to seasonal-factor revisions, so the origin dating of Section 2.3 concerns availability timing rather than literal real-time vintages.

Table A.1: Variables, Sources, and Transformations

Symbol	Series	Source	Frequency and treatment
π_t	CPI for All Urban Consumers, all items, not seasonally adjusted	FRED `CPIAUCNS`	Monthly, year-over-year percent change
π_t^{core}	CPI for All Urban Consumers, all items less food and energy, not seasonally adjusted	FRED `CPILFENS`	Quarterly, year-over-year percent change; core classification of Appendix B only
$\mathbb{E}_t \pi_{t+12}$ (MSC)	Median expected price change, next 12 months	University of Michigan, via FRED `MICH`	Monthly
$\mathbb{E}_t \pi_{t+12}$ (SPF)	Median one-year-ahead CPI inflation forecast	Federal Reserve Bank of Philadelphia, SPF `Inflation` file, variable `INFCPI1YR`	Quarterly, dated at the final month of the survey quarter
$\mathbb{E}_t \pi_{t+12}$ (Livingston)	Median forecast CPI levels	Federal Reserve Bank of Philadelphia, Livingston `medians` file, CPI sheet	Semiannual, annualized from forecast levels
i_t	Three-month Treasury bill, secondary market rate	FRED `TB3MS`	Monthly, percent per annum
$\mathcal{U}_t$	Civilian unemployment rate	FRED `UNRATE`	Monthly, percent, seasonally adjusted
Regime_t	Volatile-regime indicator	Authors' estimation of equation (2)	Quarterly, cutoff 0.5 on smoothed probabilities

Notes: The quarterly inflation series behind equation (2) and Appendix B is the year-over-year rate of each quarter's final month rather than a quarterly average, and the core series of Appendix B is built the same way; the 2025:Q4 observation therefore uses the December 2025 rate and is unaffected by the never-published October 2025 release. The Livingston expectation follows the Federal Reserve Bank of Philadelphia's documentation. Before June 1992, it is the annualized growth from the base-month CPI B, two months before the survey month, to the twelve-month-ahead forecast level F_{12}, $100[(F_{12}/B)^{12/14} - 1]$; from June 1992, when respondents begin estimating the survey-month CPI, it is the simple growth from that estimate to F_{12}. SPF and Livingston forecasts are dated at the final month of the survey period (March, June, September, or December, and June or December), so professional observation dates coincide with MSC months. Quarterly probabilities are dated at the quarter end, and each month within a quarter inherits the quarter's smoothed and filtered probabilities, and hence any classification derived from them.

Figure A.1: Year-over-Year U.S. CPI Inflation



Notes: Monthly year-over-year CPI inflation, 1981:M9–2025:M12. Shaded bars denote NBER recessions. The one-month gap in late 2025 is the never-published October 2025 CPI. Source: U.S. Bureau of Labor Statistics, via FRED.

B Regime Validation

The Markov-switching autoregression of Section 2.2 has a state-dependent innovation variance, so it partitions any sample into a high- and a low-variance state whether or not the variance of inflation genuinely shifts. The classification cannot validate itself. We therefore test the premise two ways. Table B.1 collects procedures that assume no Markov-switching data-generating process. Its first two panels never see the regime indicator. Its last two bring the indicator in as a regressor, Panel C to ask whether it absorbs the GARCH persistence and Panel D to ask whether it predicts the variance directly. Table B.2 asks whether the classification is an energy phenomenon.

We work with the residuals of an autoregression of year-over-year CPI inflation, with lag order chosen by the Bayesian information criterion (BIC): AR(6) for the 172 usable quarters of the classification window, and AR(13) for the 507 usable months of the monthly evaluation window. Filtering matters, because year-over-year inflation is strongly autocorrelated (at quarterly sampling the twelve-month windows overlap by nine months) and because the object of interest is the innovation variance, which is $\sigma^2_{s_t}$ in equation (2). We report results for demeaned but unfiltered inflation alongside.

Which parameter is unstable? Panel A reports the Nyblom (1989) and Hansen (1992) L_c statistic, which tests constant parameters against martingale variation, with a separate statistic for each parameter of the autoregression, the innovation variance in-

cluded. None of the conditional-mean parameters, the intercept included, comes close to the 5% critical value of 0.470 at either frequency. The largest is 0.235. The variance statistic is 0.398 at quarterly frequency, significant only at the 10% level, and 1.283 at monthly frequency, significant at the 1% level. Over this sample, the conditional mean of inflation is stable and the conditional variance is not. The joint statistic rejects at neither frequency, as expected when one parameter of eight or fifteen is unstable. Spreading the test across the stable coefficients dilutes its power, so the individual statistics are the informative object.

Because the score for the variance parameter, $e_t^2 - \sigma^2$, is not a martingale difference under conditional heteroskedasticity, we also report the variance statistic normalized by a Bartlett long-run variance. The monthly statistic survives the robustification at the 1% level, 0.905 against 0.748. The quarterly statistic does not, falling to 0.326. We return to that asymmetry below.

Is there a break in the unconditional variance? Panel B reports the cumulative-sum-of-squares statistic of Inclán and Tiao (1994), abbreviated IT, together with the two corrections of Sansó et al. (2004), written κ_1^S and κ_2^S to keep them distinct from the interest-rate coefficients of equation (9). The Inclán–Tiao statistic assumes independent innovations. The κ_1^S correction rescales it by the empirical fourth moment and so tolerates excess kurtosis. The κ_2^S correction divides by a Bartlett estimate of the long-run variance of e_t^2 and is therefore robust to GARCH. The gap between IT and κ_2^S is the diagnostic for the objection that an apparent variance regime is only conditional heteroskedasticity. All three statistics have the limiting distribution $\sup_r |W^*(r)|$, with quantiles 1.224, 1.358, and 1.628 from the closed-form Kolmogorov–Smirnov expression.

At monthly frequency, the evidence survives the correction with room to spare. On AR residuals, the statistic falls from 3.751 to 2.002 under the GARCH-robust normalization and still exceeds the 1% critical value, with 1.945 under the data-driven bandwidth. At quarterly frequency, the evidence is only suggestive. The κ_2^S statistic is 1.184 at the baseline bandwidth, just below the 10% critical value, and 1.263 at the data-driven bandwidth, just above it. The Inclán–Tiao statistic, by contrast, is 2.487, significant at the 1% level. The correction for dependence in e_t^2 absorbs almost all of the quarterly evidence. Applied iteratively to the monthly residuals as the iterated-cumulative-sum-of-squares (ICSS) algorithm of Inclán and Tiao (1994), the κ_2^S test locates two change points, 2005:M3 and 2010:M12, with residual standard deviations of 0.221, 0.487, and 0.298 across the three implied segments. The quarterly procedure locates none.

Three things separate the two frequencies, and we do not read the quarterly non-

rejection as overturning the monthly rejection. The quarterly sample is 172 observations against 507. The HAC correction is proportionately heavier there. Most importantly, the cumulative-sum-of-squares family is built against a single permanent break, whereas the classification has nine transitions. The partial-sum path returns toward zero each time the variance returns to its earlier level, which is precisely the configuration in which a supremum test loses power. At the frequency at which we estimate the Markov-switching model, the model-free evidence is therefore suggestive rather than established.

Do the located dates match the model? They do, without having been told to. The longest volatile episode in the classification runs from 2004:Q2 to 2011:Q2, twenty-nine quarters. The first monthly change point, 2005:M3, falls three quarters after the transition that opens it. The second, 2010:M12, falls three quarters before the transition that closes it. The two procedures share no parametric structure, one a smoothed-probability classification from a maximum-likelihood Markov-switching autoregression, the other a nonparametric sequential search on squared residuals.

Break or GARCH? Panel C runs the horse race directly, following Lamoureux and Lastrapes (1990). A Bollerslev (1986) GARCH(1,1) fitted to the monthly AR residuals returns $\hat{\alpha} + \hat{\beta} = 0.989$, the near-integrated persistence that neglected variance shifts are known to manufacture. Adding the ICSS segment dummies to the conditional-variance equation raises the log likelihood by 11.9 points, with $p < 0.001$ on two degrees of freedom, and cuts persistence to 0.630. Substituting the Markov-switching volatile dummy cuts it to 0.278. We do not lean on the second number, because that dummy comes from a model fitted to the same series under the same alternative. Its likelihood-ratio statistic is therefore mechanically favored and is not a valid test. The ICSS row is the clean one and delivers the same answer more conservatively. The data contain discrete variance shifts, not GARCH persistence alone.

Does the classification predict variance? Panel D asks the complementary question, and it is the one that matters for the rest of the paper: does the regime label carry information about the size of inflation innovations? Regressing squared AR residuals on the volatile dummy gives $t = 4.95$ at monthly frequency and $t = 3.62$ at quarterly frequency, with residual standard deviations 1.82 and 2.37 times larger in the volatile state. The result survives restriction to the 2011-onward subsample, with $t = 2.14$ at monthly and $t = 2.53$ at quarterly frequency. That restriction matters, because the ICSS procedure does not isolate the post-2021 surge as a separate variance segment. The AR(13) filter absorbs much of 2021–2023 as a level movement, so the third ICSS segment spans 2011–2024 as a unit. Panel D, not Panel B, supports the classification of the post-2021 surge. Not every volatile episode is independently corroborated.

Is the volatile regime an energy phenomenon? The classification comes from headline inflation, and several of the episodes it selects are oil-price events. We therefore re-estimate equation (2) on core inflation, which excludes food and energy, over the same sample. Core inflation is quieter. Its stable-state innovation variance is 0.055 against 0.211 for headline inflation, its stable state is more persistent at $\widehat{p}_{11} = 0.991$, and it yields 24 volatile quarters against 64. The two classifications agree in 134 of the 178 quarters, and the core volatile quarters are nearly a subset of the headline ones, 22 of the 24 being volatile under both. Table B.2 reports the comparison.

Where the two part company is the informative part. Both call the early-1980s disinflation volatile, from 1981:Q3 to 1983:Q4, and both call the post-pandemic surge volatile, from 2020:Q4 to 2023:Q3. The 42 quarters that only headline inflation calls volatile are the 1986 oil-price collapse, the 1990 spike, the long commodity run-up through the financial crisis from 2004:Q2 to 2011:Q2, and 1984:Q1, the quarter in which the two classifications diverge at the end of the disinflation. The two quarters that only core calls volatile, 2020:Q2–2020:Q3, date the onset of the pandemic surge two quarters earlier than headline does. Energy therefore accounts for much of the volatile regime, but for neither of the two episodes that dominate the evaluation window. We retain the headline classification because the three surveys forecast headline inflation, so the regime belongs on the variable being forecast, but a reader who prefers a core-based reading should discount the middle of the sample rather than its ends.

What this appendix does and does not establish. It establishes three things. Over this sample, inflation has an unstable innovation variance and a stable conditional mean. The instability is not an artifact of conditional heteroskedasticity at monthly frequency, and the quarterly evidence on that question is suggestive rather than established. Finally, a nonparametric search recovers both ends of the model's longest volatile episode. It does not establish that two states is the right number, that a first-order Markov chain is the right transition mechanism, that any particular quarter is correctly classified, or that the volatile episodes in the middle of the sample are anything other than energy-price events. Those remain modeling choices. Section 6.2 reports how the results change under alternative classifications built from the same model.

Table B.1: Variance Instability in Realized Inflation

Panel A: Nyblom–Hansen L_c parameter-constancy test

				Individual statistics		
Series	n	m	Joint L_c	$\max_j L_{\beta_j}$	L_{σ^2}	L_{σ^2}, HAC
Quarterly, AR(6)	172	8	1.172	0.230	0.398*	0.326
Monthly, AR(13)	507	15	3.143	0.235	1.283***	0.905***

Panel B: cumulative-sum-of-squares tests for a variance break

Series	IT	κ_1^S	κ_2^S	κ_2^S, NW bw	Break date
Quarterly, demeaned	2.771***	1.854***	1.024	1.179	2008:Q1
Quarterly, AR residuals	2.487***	1.307*	1.184	1.263*	2005:Q2
Monthly, demeaned	4.945***	3.508***	1.520**	1.256*	2008:M5
Monthly, AR residuals	3.751***	2.383***	2.002***	1.945***	2005:M3

Panel C: GARCH(1,1) persistence horse race, monthly AR(13) residuals

Conditional-variance specification	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha}+\hat{\beta}$	LR	df	p
GARCH(1,1)	0.073	0.916	0.989	—	—	—
+ ICSS segment dummies	0.088	0.543	0.630	23.80	2	< 0.001
+ MS volatile dummy	0.169	0.109	0.278	25.24	1	< 0.001

Panel D: HAC regression of squared AR residuals on the volatile dummy

Sample	n	n volatile	$\hat{v}$	t	p	sd ratio
Monthly, full sample	507	177	0.111	4.95	< 0.001	1.82
Monthly, 2011 onward	168	42	0.078	2.14	0.033	1.37
Quarterly, full sample	172	58	0.959	3.62	< 0.001	2.37
Quarterly, 2011 onward	60	14	0.978	2.53	0.014	2.05

Notes: Year-over-year CPI inflation; the quarterly series is the classification window (1981:Q3–2025:Q4), the monthly series the evaluation window (1981:M9–2024:M12). $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Panel A. m counts parameters including σ^2; $\max_j L_{\beta_j}$ is the largest statistic across the intercept and the autoregressive coefficients. Hansen (1992) critical values at 10%/5%/1%: 0.353/0.470/0.748 for one parameter, 1.89/2.11/2.59 for $m = 8$, 3.26/3.54/4.07 for $m = 15$. The last column normalizes the variance score by a Bartlett long-run variance rather than by its outer product, at the Panel B baseline bandwidth.

Panel B. IT is the Inclán and Tiao (1994) statistic; κ_1^S and κ_2^S are the Sansó et al. (2004) corrections robust to excess kurtosis and to conditional heteroskedasticity, respectively. Critical values at 10%/5%/1%: 1.224/1.358/1.628. Bandwidths are $\lfloor 4(T/100)^{2/9} \rfloor$ (four quarterly, five monthly), and in the "NW bw" column the automatic rule of Newey and West (1994); the break date is the argmax of the κ_2^S path. The iterated cumulative-sum-of-squares (ICSS) search behind Panel C applies κ_2^S recursively to the AR residuals by binary segmentation at the 5% level, with a minimum segment length of 24 months, and without the refinement pass of the original algorithm.

Panel C. Gaussian quasi-maximum likelihood with $h_t = \omega + z_t'\delta + \alpha e_{t-1}^2 + \beta h_{t-1}$, where z_t holds the row's dummies, and h_1 set to the sample variance; log likelihoods -63.99, -52.09, and -51.37. ICSS dummies mark the change points 2005:M3 and 2010:M12; the Markov-switching (MS) dummy is the baseline Regime_t. LR is the likelihood ratio against the first row; the MS-dummy LR is not a valid test, because that dummy comes from a model fitted to the same series.

Panel D. $\hat{v}$ is the coefficient on the baseline Regime_t, with Newey–West standard errors using VAR(1) prewhitening and the automatic bandwidth of Newey and West (1994); "sd ratio" is the volatile-to-stable ratio of residual standard deviations.

Table B.2: Headline and Core Classifications Compared

	Core stable	Core volatile	Total
Headline stable	112	2	114
Headline volatile	42	22	64
Total	154	24	178

Notes: Quarters of the 1981:Q3–2025:Q4 classification window, with the 1981:Q2 observation serving as the conditioning value. The headline classification is the one used throughout the paper. The core classification uses year-over-year inflation in the consumer price index for all items less food and energy, not seasonally adjusted, built to quarterly frequency the same way and estimated with the same specification and starting values, with volatile quarters classified by the same 0.5 cutoff on smoothed probabilities. Its estimates include a volatile-state innovation variance of 1.389, long-run means of 2.101 and 4.940, and a volatile-state transition probability of 0.901.

C Accuracy Test Specifications and Diagnostics

Section 4 concludes that neither group can be shown to be more accurate. This appendix sets out the two tests behind that conclusion in full, then asks what happens below the 95% convention and which direction the data favor. All tests use the common-date loss differences of equation (4).

Equal accuracy conditional on origin-dated information. The conditional predictive ability test of Giacomini and White (2006) asks whether anything known at the origin predicts which survey will be more accurate. The test treats forecasts as primitives rather than as products of estimated models, the relevant case for survey expectations. The null is $\mathbb{E}[d_t \mid \mathcal{F}_t] = 0$. We regress d_t on a vector of instruments z_t and compute the Wald statistic for the null that all coefficients are zero:

$$W \;=\; \hat{\beta}' \hat{V}^{-1} \hat{\beta}, \tag{C.1}$$

asymptotically χ^2 with $\dim(z_t)$ degrees of freedom, where $\hat{V}$ is a HAC estimate of the covariance of $\hat{\beta}$. We use two instrument sets, $\{1,\ \text{Regime}_t\}$ and $\{1,\ \text{Regime}_t,\ d_{t-s}\}$, with d_{t-s} the most recent loss difference verifiable at the origin.

The theory requires $z_t \in \mathcal{F}_t$, which is why Table 4 reports every test with the baseline indicator and with the filtered indicator $\text{Regime}_{f,t}$. Section 4.3 explains the distinction and reads the results. The note to Table 4 details the bootstrap complement, and Section 4.3 reports a third variant, the filtered indicator dated one quarter before the origin.

Equal accuracy at every point in the sample. A brief episode of significant dominance can still hide in a full-sample or full-regime mean. The fluctuation test of Giacomini and Rossi (2010) checks for such an episode directly, computing the standardized

rolling statistic:

$$\Phi_t \;=\; \hat{\sigma}^{-1} m^{-1/2} \sum_{j=t-m+1}^{t} d_j, \tag{C.2}$$

and comparing its entire path against critical values valid for the whole path under equal predictive ability. We use windows of $\mu = 0.3$ and 0.2 of each sample, $m = 52$ and 35 quarters for the MSC–SPF comparison and $m = 26$ and 17 half-years for MSC–Livingston, with $\hat{\sigma}^2$ the full-sample prewhitened Newey–West long-run variance at the one-year bandwidth. For the moving-block bootstrap bands, we demean the loss difference, resample it in one-year blocks, recompute the entire standardized path on each of 4,999 samples, and take the 95th percentile of the path maximum as the band, with the path-maximum p-value the exceedance share.

Figure C.1 plots all eight paths, each comparison under each loss at both window lengths. Under squared loss, the MSC–SPF path reaches 1.46 at $\mu = 0.3$ and 1.78 at $\mu = 0.2$, with a minimum of -1.02 in the shorter windows ending in 2024, against Giacomini–Rossi critical values of 3.01 and 3.18 and tighter moving-block bootstrap bands of 2.65 and 2.86. The bootstrap p-values for the path maximum are 0.651 and 0.562. The MSC–Livingston path peaks at 1.92 and 2.33, against bands of 2.94 and 3.22, with p-values of 0.456 and 0.354.[16]

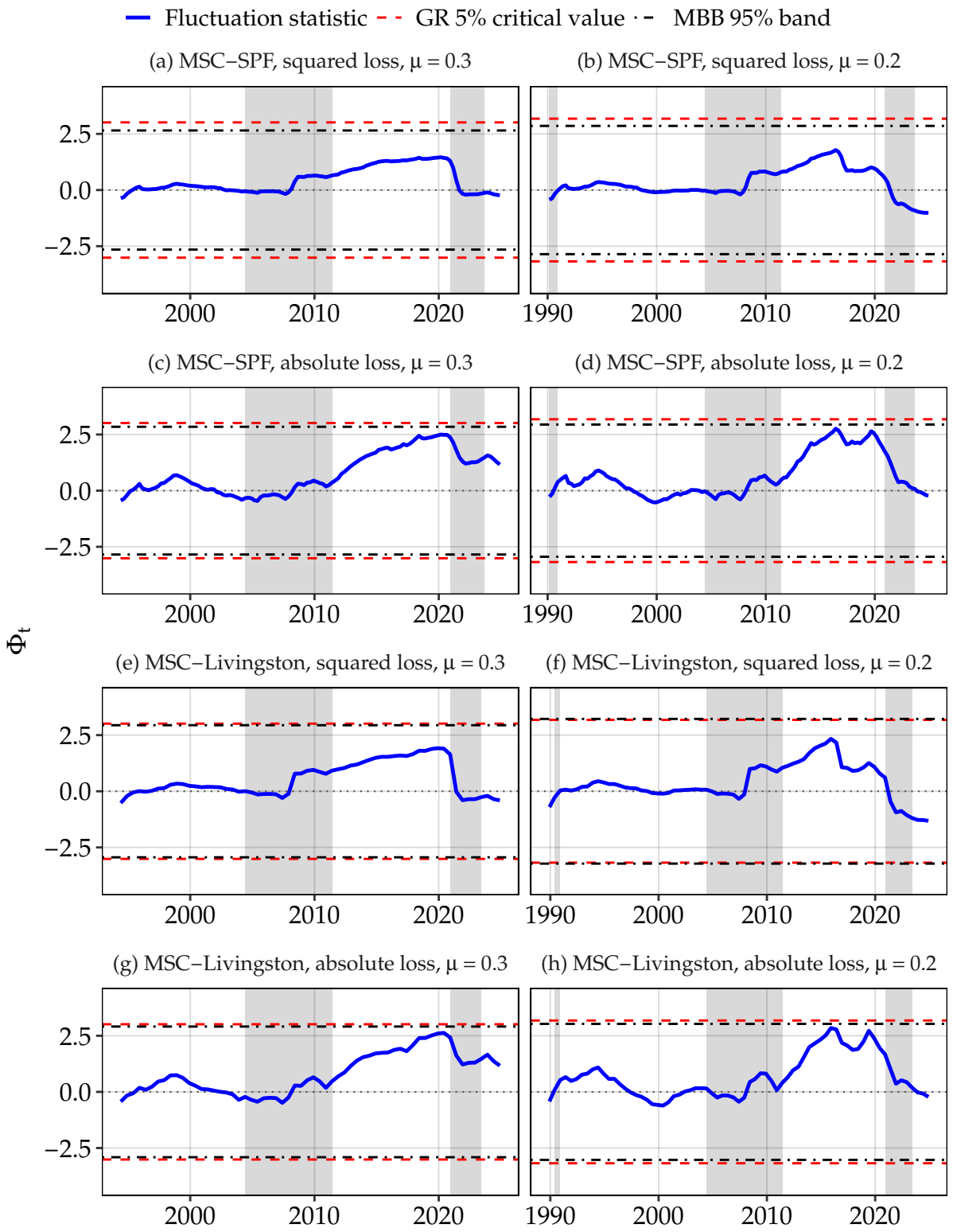
Absolute loss brings the paths closer to the bands without crossing them. The MSC–SPF maximum rises to 2.50 at $\mu = 0.3$ and 2.76 at $\mu = 0.2$, against bands of 2.84 and 2.94, and the MSC–Livingston maximum to 2.63 and 2.84, against bands of 2.91 and 3.03. The path-maximum bootstrap p-values are 0.127, 0.091, 0.102, and 0.080. All four peaks sit in windows that end between 2015 and 2020, dominated by the stable 2010s. This is the signal Table C.1 picks up below the 95% convention, a stable-regime professional advantage under absolute loss, and at its strongest it still stays inside every band.

Probing the answer. An answer as uniform as Section 4.2's invites two questions, one about the 95% convention and one about direction. Table C.1 collects the statistics behind both.

What happens below the 95% convention? At the 90% level, one of the twelve intervals in Table C.1 moves off zero. It belongs to the MSC–Livingston absolute-loss difference in the stable regime, whose 95% interval of $[-0.022,\ 0.356]$ in Table 3 tightens to $[0.005,\ 0.327]$. The interval moves off zero only at the shortest admissible block length. Blocks of three, four, or six half-years return it to containing zero at both levels. The

[16] The conclusion is also insensitive to the variance estimator. Under the non-prewhitened Bartlett kernel used in the reference implementation of the test, the MSC–SPF path scales up, but at $\mu = 0.3$ its maximum, 2.25, remains inside the critical values.

Figure C.1: Giacomini–Rossi Fluctuation Tests



Notes: Each panel plots the statistic Φ_t of equation (C.2) for the loss difference d_t, the MSC loss minus the professional loss, over rolling windows dated at their end; negative values favor households. Windows span $\mu = 0.3$ and 0.2 of the common dates, the 174 quarters of 1981:Q3–2024:Q4 for MSC–SPF and the 87 half-years of 1981:H2–2024:H2 for MSC–Livingston, so $m = 52$ and 35 quarters and 26 and 17 half-years. Dashed horizontal lines are the two-sided 5% critical values of Giacomini and Rossi (2010); dash-dotted lines are moving-block bootstrap (MBB) 95% bands for the path maximum of $|\Phi_t|$ (one-year blocks, 4,999 replications). Shaded areas mark periods classified as volatile by the baseline indicator of equation (3).

Table C.1: Loss Differences: 90% Intervals and One-Sided Tail Areas

Comparison	Full sample	Stable regime	Volatile regime
MSC − SPF, squared loss	[−0.783, 0.982] 0.411 / 0.358	[−0.086, 0.907] 0.086 / 0.137	[−2.634, 1.794] 0.378 / 0.443
MSC − SPF, absolute loss	[−0.063, 0.272] 0.151 / 0.136	[−0.008, 0.335] 0.058 / 0.106	[−0.335, 0.321] 0.511 / 0.615
MSC − Livingston, squared loss	[−0.837, 0.976] 0.436 / 0.388	[−0.045, 0.790] 0.078 / 0.113	[−2.814, 1.948] 0.374 / 0.422
MSC − Livingston, absolute loss	[−0.072, 0.267] 0.165 / 0.147	[0.005, 0.327] [a] 0.044 / 0.083	[−0.373, 0.330] 0.473 / 0.544

Notes: Cells correspond one for one to the mean loss differences of Table 3. Brackets: the 90% moving-block bootstrap percentile interval, from the same 9,999 replications as the 95% intervals of Table 3 (one-year blocks: four quarters for MSC–SPF, two half-years for MSC–Livingston). The two numbers beneath each interval are one-sided bootstrap tail areas: the share of the 9,999 bootstrap means that fall on the opposite side of zero from the point estimate, so a small value is directional support for the sign of the point estimate. The first tail area uses the one-year block length, the second a two-year block length (eight quarters / four half-years). [a]The only interval that excludes zero at the 90% level. It contains zero at semiannual block lengths of three, four, and six half-years.

Newey–West p-value is 0.26 to 0.28 at every bandwidth from two to six. That series also shows autocorrelation beyond the order the twelve-month overlap implies, with a data-driven bandwidth of 4.92 against the fixed two, which makes a longer block the better choice.

Which direction do the data favor? The one-sided bootstrap tail areas of Table C.1 speak to direction where the two-sided intervals cannot, and they are asymmetric across regimes. At the baseline block length, the tail area opposite the point estimate is 0.044 to 0.086 in all four stable-regime comparisons and 0.37 to 0.51 in all four volatile-regime comparisons. Lengthening the blocks to two years raises both, to 0.08–0.14 and 0.42–0.62. These tail areas measure directional support for the sign of the point estimate, not size under a null, so their levels are not p-values. The stable–volatile gap, a factor of three to eleven, persists at every block length we consider. What little directional support the data contain sits with the professionals in the stable regime.

D Additional Expectation-Formation Results

This appendix supports Section 5 with three exercises. Tables D.1 and D.2 remove the regime terms from every specification, to record what the familiar full-sample tests deliver and so to show which results exist only under regime conditioning. Table D.3 replaces the distributed lag with a single lag of inflation, to show that the household regime shift and the signs of the professional shifts do not depend on the lag polynomial. Table D.4 estimates a revision-based alternative to the adaptive regression, to document the qualification reported in Section 5.2.

D.1 Results without regime interactions

Is anything in Section 5 visible without the regime split? Tables D.1 and D.2 answer by re-estimating the four rationality tests and the adaptive-expectations regression with no regime terms.

Without the interactions, the tests collapse into familiar full-sample results. Panel A shows no significant bias for any survey. In Panel B, the own-forecast efficiency test does not reject for the MSC but rejects for the SPF and the Livingston Survey. The bootstrap qualifies this panel twice. It places the MSC slope's interval strictly below zero even though the joint test does not reject on either route, and it leaves the SPF rejection unconfirmed and confirms the Livingston rejection only at the 10% level. Panel C shows no significant full-sample error persistence, and in Panel D the macroeconomic-information test does not reject for any survey. What the regime split adds is location. Only with it can we place the professional efficiency failures in the volatile regime, and without it the household stable-regime departures, namely persistent errors and unexploited macroeconomic information, disappear entirely.

Without regime terms, the cumulative loading on realized inflation is significant at the 1% level for every survey, as Table D.2 reports. The macroeconomic controls stay jointly significant for all three surveys, so no survey follows a purely backward-looking rule.

D.2 Adaptive expectations with a single inflation lag

Do the regime shifts in the loadings depend on the twelve-lag polynomial? Table D.3 answers by re-estimating the adaptive-expectations regression of Table 9 with a single lag of realized inflation, π_{t-1}, in place of the twelve monthly lags of equation (10). The answer is no. The regime interaction remains positive and significant for the MSC and negative for both professional surveys, significantly so for the Livingston Survey.

D.3 An alternative test using forecast-error revisions

Does the regime dependence of the loadings on recent inflation extend to the speed of error correction? Following Pfajfar and Santoro (2010), we answer with the alternative specification of Table D.4, in which forecast updating is captured by the latest verifiable forecast error:

$$\mathbb{E}_t[\pi_{t+12}] = \mathbb{E}_{t-s}[\pi_{t-s+12}] + (\zeta_1 + \zeta_2 \,\text{Regime}_t)\big(\pi_{t-s+12} - \mathbb{E}_{t-s}[\pi_{t-s+12}]\big) + \varepsilon_t, \quad \text{(D.1)}$$

Table D.1: Tests of Forecast Rationality without Regime Terms

	MSC	SPF	Livingston
Panel A: unbiasedness			
α	-0.307 (0.393) $[-0.674, 0.075]$	-0.088 (0.391) $[-0.445, 0.360]$	-0.065 (0.311) $[-0.446, 0.367]$
Panel B: own-forecast efficiency			
β	-0.482 (0.607) $[-0.909, -0.024]$	-0.536^{**} (0.250) $[-0.851, -0.181]$	-0.533^{***} (0.197) $[-0.866, -0.139]$
α	1.225 (1.537) $[-0.269, 2.661]$	1.497 (0.990) $[0.436, 2.564]$	1.491^{*} (0.783) $[0.370, 2.566]$
L6: Reject H_0: $\alpha = \beta = 0$?	F=0.320 p=0.727 p_{MBB}=0.897	F=3.498 p=0.032 p_{MBB}=0.104	F=5.324 p=0.007 p_{MBB}=0.058
Panel C: error persistence			
β	0.057 (0.198) $[-0.151, 0.267]$	0.183 (0.203) $[-0.015, 0.406]$	0.208 (0.157) $[-0.014, 0.465]$
α	-0.289 (0.396) $[-0.658, 0.096]$	0.005 (0.295) $[-0.327, 0.399]$	-0.045 (0.239) $[-0.406, 0.370]$
Panel D: use of macroeconomic information			
α	1.435 (1.329) $[-0.310, 3.161]$	1.367^{**} (0.638) $[-0.170, 2.845]$	1.221^{**} (0.569) $[-0.296, 2.712]$
β	-0.943 (0.634) $[-1.468, -0.403]$	-0.601^{*} (0.327) $[-1.412, 0.259]$	-0.665^{***} (0.244) $[-1.468, 0.166]$
γ	0.188 (0.160) $[-0.084, 0.474]$	0.114 (0.229) $[-0.176, 0.410]$	0.144 (0.178) $[-0.140, 0.441]$
κ	0.071 (0.190) $[-0.074, 0.210]$	-0.027 (0.163) $[-0.280, 0.211]$	-0.016 (0.143) $[-0.269, 0.225]$
δ	0.070 (0.105) $[-0.121, 0.277]$	0.014 (0.102) $[-0.207, 0.246]$	0.047 (0.097) $[-0.171, 0.276]$
L7: Reject H_0: $\gamma = \kappa = \delta = 0$?	F=1.062 p=0.365 p_{MBB}=0.359	F=0.107 p=0.956 p_{MBB}=0.962	F=0.350 p=0.789 p_{MBB}=0.879

Notes: Each panel estimates the corresponding specification of Section 2.3, equations (6)–(9), with all regime terms (the regime dummy and its interactions) removed. Samples: MSC 519 monthly (1981:M9–2024:M12), SPF 174 quarterly, Livingston 87 semiannual. The October 2024 MSC forecast error is unavailable because the October 2025 CPI was never published. Panel C uses 170 SPF observations (the lagged own error is unavailable at the start of the SPF sample). Prewhitened Newey–West standard errors in parentheses (bandwidth of one year: 12, 4, and 2 lags). Brackets: residual moving-block bootstrap 95% percentile intervals from 4,999 replications; p_{MBB} beneath L6 and L7: null-imposed residual-MBB p-values from 1,999 replications. Stars from Newey–West t-tests: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table D.2: Adaptive Expectations without Regime Terms

	MSC	SPF	Livingston
$\beta(1)$	0.404^{***}	0.225^{***}	0.169^{***}
	(0.023)	(0.038)	(0.042)
	[0.345, 0.459]	[0.149, 0.302]	[0.084, 0.254]
α	1.880^{***}	0.471	0.511
	(0.146)	(0.382)	(0.348)
	[1.599, 2.174]	[0.076, 0.859]	[0.070, 0.929]
λ_1	-0.071^{***}	0.196	0.177
	(0.020)	(0.171)	(0.180)
	[−0.136, 0.001]	[−0.087, 0.473]	[−0.218, 0.568]
λ_2	0.116^{***}	−0.047	−0.018
	(0.022)	(0.138)	(0.148)
	[0.042, 0.182]	[−0.298, 0.216]	[−0.384, 0.348]
θ_1	−0.025	0.330^{*}	0.580^{**}
	(0.118)	(0.170)	(0.229)
	[−0.190, 0.138]	[0.047, 0.596]	[0.092, 1.070]
θ_2	−0.029	−0.084	−0.321
	(0.111)	(0.156)	(0.224)
	[−0.192, 0.131]	[−0.351, 0.199]	[−0.806, 0.161]
L8: Reject H_0: $\lambda_1 = \lambda_2 = \theta_1 = \theta_2 = 0$?	F=11.427	F=39.630	F=25.194
	p<0.001	p<0.001	p<0.001
	p_{MBB}=0.011	p_{MBB}=0.001	p_{MBB}=0.001
R^2	0.672	0.886	0.860
Adjusted R^2	0.661	0.875	0.829

Notes: Estimates of equation (10) with the regime interaction removed. Samples: MSC 520 monthly (1981:M9–2024:M12), SPF 174 quarterly, Livingston 87 semiannual. $\beta(1)$ is the cumulative loading, the sum of the twelve monthly inflation-lag coefficients; other symbols as in Table 9. Prewhitened Newey–West standard errors in parentheses (bandwidth of one year: 12, 4, and 2 lags). Brackets: residual moving-block bootstrap 95% percentile intervals from 4,999 replications; p_{MBB} beneath L8: null-imposed residual-MBB p-value from 1,999 replications. Stars from Newey–West t-tests: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table D.3: Adaptive Expectations with a Single Inflation Lag

	MSC	SPF	Livingston
β_1	0.263^{***}	0.250^{***}	0.257^{***}
	(0.038)	(0.073)	(0.057)
	[0.196, 0.328]	[0.166, 0.335]	[0.154, 0.359]
β_2	0.093^{***}	-0.030	-0.070^{**}
	(0.027)	(0.046)	(0.034)
	[0.049, 0.138]	[−0.088, 0.028]	[−0.138, −0.002]
$\beta_1 + \beta_2$	0.356^{***}	0.220^{***}	0.186^{***}
	(0.024)	(0.041)	(0.039)
	[0.310, 0.401]	[0.161, 0.281]	[0.121, 0.255]
α	2.085^{***}	0.380	0.354
	(0.136)	(0.443)	(0.389)
	[1.795, 2.373]	[−0.028, 0.784]	[−0.109, 0.798]
λ_1	-0.069^{***}	0.215	0.156
	(0.017)	(0.152)	(0.148)
	[−0.137, 0.006]	[−0.062, 0.482]	[−0.213, 0.509]
λ_2	0.112^{***}	-0.055	0.012
	(0.022)	(0.125)	(0.130)
	[0.037, 0.181]	[−0.301, 0.198]	[−0.322, 0.356]
θ_1	-0.069	0.322^{*}	0.735^{***}
	(0.115)	(0.163)	(0.225)
	[−0.225, 0.089]	[0.056, 0.574]	[0.258, 1.227]
θ_2	0.046	-0.076	-0.495^{**}
	(0.111)	(0.155)	(0.220)
	[−0.110, 0.200]	[−0.327, 0.188]	[−0.985, −0.016]
R^2	0.675	0.884	0.856
Adjusted R^2	0.671	0.880	0.846

Notes: Estimates of equation (10) with both lag polynomials collapsed to scalars, $\mathbb{E}_t[\pi_{t+h}] = \alpha + (\beta_1 + \beta_2\,\text{Regime}_t)\,\pi_{t-1} + \text{controls}$, where π_{t-1} is the one-month lag for all three surveys and $\beta_1 + \beta_2$ is the volatile-regime loading. Other symbols as in Table 9. Samples: MSC 520 monthly (1981:M9–2024:M12), SPF 174 quarterly, Livingston 87 semiannual. Prewhitened Newey–West standard errors in parentheses (bandwidth of one year: 12, 4, and 2 lags). Brackets: residual moving-block bootstrap 95% percentile intervals from 4,999 replications. Stars from Newey–West t-tests: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

where ζ_1 is the baseline updating speed in the stable regime and ζ_2 the change in the volatile regime. We impose the unit coefficient on the earlier forecast, so we regress the forecast revision on the interacted verifiable error, without an intercept. The revision lag s is the smallest multiple of the survey's frequency such that the earlier forecast's target period precedes t, so that the forecast is verifiable when the time-t forecast is formed. The lag is thirteen months for the monthly MSC (so the error term is $\pi_{t-1} - \mathbb{E}_{t-13}\pi_{t-1}$), fifteen months for the quarterly SPF, and eighteen months for the semiannual Livingston Survey.

We detect no such extension. In the stable regime, the updating speed ζ_1 is positive and significant for the MSC and the SPF, and positive but insignificant for the Livingston Survey. In the volatile regime, the sums $\zeta_1 + \zeta_2$ are positive and significant for all three surveys. The household speed rises from 0.261 to 0.398, the SPF speed moves only from 0.240 to 0.245, and the Livingston speed rises from 0.125 to 0.241. The household rise points the same way as in Table 9, where households weight recent inflation more heavily once it turns volatile. The shifts ζ_2 are statistically insignificant for every survey, so the revision test cannot establish any regime dependence. The two findings do not contradict each other, because the two regressions measure different objects. The adaptive regression asks how the forecast level weights recently realized inflation. The revision test asks how fast the forecast corrects its latest verifiable error. A forecaster can change the weight on recent inflation when inflation turns volatile and still correct verifiable errors at an unchanged speed. The revision test is also coarser than the adaptive regression, since it condenses updating into a single verifiable error in place of the distributed lag.

Table D.4: Forecast-Error-Revision Test of Adaptive Expectations

	MSC	SPF	Livingston
ζ_1	0.261***	0.240***	0.125
	(0.061)	(0.073)	(0.077)
	[0.126, 0.434]	[0.144, 0.408]	[−0.071, 0.473]
ζ_2	0.138	0.005	0.116
	(0.088)	(0.085)	(0.131)
	[−0.064, 0.304]	[−0.179, 0.111]	[−0.272, 0.328]
$\zeta_1 + \zeta_2$	0.398***	0.245***	0.241**
	(0.065)	(0.049)	(0.103)
	[0.290, 0.498]	[0.177, 0.304]	[0.072, 0.359]
R^2	0.347	0.436	0.172
Adjusted R^2	0.345	0.429	0.153

Notes: Estimates of equation (D.1), a regression of the forecast revision on the latest verifiable forecast error without an intercept, with revision lags s of 13, 15, and 18 months; R^2 is uncentered. ζ_1 is the stable-regime updating speed and $\zeta_1 + \zeta_2$ the volatile-regime speed. Samples: MSC 520 monthly (1981:M9–2024:M12), SPF 169 quarterly, Livingston 87 semiannual. The MSC and Livingston samples are unchanged because both surveys supply pre-window forecasts for the lagged term; the SPF's CPI question begins in 1981:Q3, so its fifteen-month lag costs the first five quarters. Prewhitened Newey–West standard errors in parentheses (bandwidth of one year: 12, 4, and 2 lags). Brackets: residual moving-block bootstrap 95% percentile intervals from 4,999 replications. Stars from Newey–West t-tests: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

E Estimator Diagnostics and Bootstrap Procedures

This appendix supports the inference in Sections 6.3 and 6.4. Appendix E.1 documents the serial correlation in the loss difference, which also underlies the effective sample sizes of Section 4.4, and Appendix E.2 reports the prewhitening and small-sample checks. Appendix E.3 specifies the residual moving-block bootstrap used throughout the paper, and Appendix E.4 details the parametric bootstrap of Section 6.4.

E.1 Serial correlation in the loss difference

The twelve-month horizon makes d_t a moving average of order $q = h/\text{freq} - 1$ even when both forecasts are optimal: $q = 3$ for the quarterly MSC–SPF comparison and $q = 1$ for the semiannual MSC–Livingston comparison. Autocorrelation up to lag q is mechanical. What matters for inference is autocorrelation beyond lag q, which the overlap does not explain. Table E.1 reports the diagnostic.

Panel A shows that the serial correlation present is very nearly what the overlap predicts. In both series, the excess-autocorrelation test does not reject, which is the pattern Section 6.3 relies on. The variance inflation in Panel A is nonetheless substantial for the quarterly comparison, a factor of 6.7 under squared loss over the full sample.

Panel B converts that variance inflation into the quantity that matters for power. The effective sample sizes are small, most severely for the MSC–SPF squared-loss dif-

ference in the volatile regime, where 64 nominal observations carry the information of about ten independent ones.

E.2 Prewhitening and small-sample corrections

Prewhitening. The Andrews–Monahan prefilter is worth its variance cost only when the score series is persistent. Our score series is persistent. The first-order autocorrelation of the loss difference is 0.742 for the quarterly comparison, with a t statistic of 14.6, and 0.300 for the semiannual one, with a t statistic of 3.0. The quarterly series is squarely in the case prewhitening was designed for. The semiannual series is persistent but only moderately so, and there prewhitening moves the standard error only modestly. In both series, prewhitening raises the standard error relative to the textbook estimator, so it is the conservative choice.

Small-sample corrections. Three conventions are at issue. The baseline draws p-values from a t distribution with residual degrees of freedom rather than from the normal, and it reports moving-block bootstrap intervals or p-values alongside every survey evaluation except the common-date check of Table 10. The bootstrap is itself a small-sample device that does not rely on asymptotic approximation. The baseline does not apply the $n/(n-k)$ scaling of the covariance estimator, and it does not apply

Table E.1: Serial Correlation in the Loss Difference d_t

Panel A: autocorrelation structure, full sample

			Autocorrelation			p-value			
Result	n	q	ρ_1	ρ_2	ρ_3	LB(8)	Excess	VIF	$\widehat{bw}$
MSC−SPF loss difference	174	3	0.74	0.41	0.17	< 0.001	0.442	6.74	7.67
MSC−Livingston loss difference	87	1	0.30	0.02	0.05	0.351	0.230	1.64	2.96

Panel B: effective sample size by regime

	Stable regime			Volatile regime		
Result	n	VIF	eff. n	n	VIF	eff. n
MSC−SPF loss difference	110	4.80	23	64	6.47	10
MSC−Livingston loss difference	55	2.08	27	32	1.43	22

Notes: d_t is the squared-loss difference of equation (4) on common dates. q is the moving-average order implied by the twelve-month overlap alone. ρ_j is the sample autocorrelation of d_t at lag j; the 5% white-noise band is ± 0.149 for $n = 174$ and ± 0.210 for $n = 87$. LB(8) is the Ljung–Box p-value at eight lags, computed on the residuals from the regime-mean regression, the series that enters the inference. "Excess" is the p-value of a joint test that the coefficients on lags $q+1$ through $q+4$ are zero in a regression of d_t on an intercept and those four lags only; the mechanical MA(q) correlation at lags $\leq q$ remains in the error and is absorbed by the prewhitened Newey–West covariance at the baseline bandwidth. VIF is the variance inflation factor, the ratio of the prewhitened Newey–West variance of the mean to its i.i.d. counterpart; effective n is the nominal sample size divided by the unrounded VIF and then rounded. In Panel B, the numerator is the regime-mean coefficient variance from the full-sample regime regression that underlies Table 3, at the baseline bandwidth, and the denominator is the regime's own variance divided by its n. $\widehat{bw}$ is the data-driven bandwidth of Newey and West (1994), for comparison with the fixed one-year bandwidths of four and two used in the baseline.

Table E.2: Estimator and Small-Sample Sensitivity of the Loss-Difference Tests

Result	Sample	$\bar{d}$	p-value		
			Baseline	$n/(n-k)$ scaled	HLN corrected
MSC−SPF loss difference	Full sample	0.038	0.965	0.965	0.953
	Stable regime	0.405	0.324	0.327	0.264
	Volatile regime	−0.592	0.787	0.788	0.714
MSC−Livingston loss difference	Full sample	0.009	0.988	0.988	0.988
	Stable regime	0.357	0.270	0.276	0.224
	Volatile regime	−0.588	0.704	0.707	0.722

Notes: Squared-loss differences d_t of equation (4); $\bar{d}$ reproduces Table 3. "Baseline" is the prewhitened Newey–West estimator at the fixed one-year bandwidth without finite-sample scaling, referred to t_{n-k} with k the number of regressors, as used throughout the paper. "$n/(n-k)$ scaled" applies the $n/(n-k)$ finite-sample scaling to the same estimator. "HLN corrected" applies the Harvey–Leybourne–Newbold correction of equation (E.1), with $h = 4$ quarters or 2 half-years, to a Diebold–Mariano statistic built from the truncated variance estimator $\hat{\gamma}_0 + 2\sum_{j=1}^{h-1}\hat{\gamma}_j$ of Harvey et al. (1997), referred to t_{n-1}.

the Harvey et al. (1997) correction to the Diebold–Mariano statistic:

$$DM^* = DM \times \sqrt{\frac{n+1-2h+h(h-1)/n}{n}} \;\sim\; t_{n-1}, \tag{E.1}$$

where h is the horizon in sampling units ($h = 4$ quarters, $h = 2$ half-years). Table E.2 adds both the $n/(n-k)$ scaling and the Harvey–Leybourne–Newbold correction. Neither changes any conclusion. The $n/(n-k)$ scaling moves p-values by at most 0.006. The Harvey–Leybourne–Newbold correction combines shrinkage factors of 0.980 and 0.983 with a truncated variance estimator smaller than the prewhitened Newey–West one, and it lowers the p-value in four of the six cells and leaves one unchanged. In the remaining cell, the MSC–Livingston volatile-regime difference, the shrinkage factor dominates the slightly smaller variance and the p-value rises by 0.018. Wherever the correction moves a p-value materially, it moves it down, so the baseline remains the more conservative convention. Even so, the smallest p-value anywhere in the table is 0.224. No cell approaches significance under any combination. The gaps that make the regime cells non-contiguous break the moving-average structure, so the correction is an approximation there. We compute the truncated autocovariances and n on the regime's observations placed side by side, with h unchanged, which if anything overstates the dependence.

E.3 Residual moving-block bootstrap for regression quantities

Because one-year-ahead forecast errors overlap, HAC inference may be inaccurate in samples of this size. We therefore run a residual moving-block bootstrap for every survey-based regression quantity reported in the paper's coefficient tables, except the common-date check of Table 10. Two independent implementations in the replication

Table E.3: Moving-Block Bootstrap Inference for Key Coefficients

Result	Estimate (s.e.)	MBB 95% interval
SPF own-forecast efficiency, stable (β_1)	-0.230 (s.e. 0.272)	$[-0.681,\ 0.274]$
Livingston own-forecast efficiency, stable (β_1)	-0.173 (s.e. 0.222)	$[-0.641,\ 0.335]$
SPF own-forecast efficiency, volatile ($\beta_1 + \beta_2$)	-0.831 (s.e. 0.288)	$[-1.223,\ -0.415]$
Livingston own-forecast efficiency, volatile ($\beta_1 + \beta_2$)	-0.871 (s.e. 0.245)	$[-1.285,\ -0.399]$
MSC cumulative loading, stable ($\beta_1(1)$)	0.305 (s.e. 0.038)	$[0.227,\ 0.382]$
SPF cumulative loading, stable ($\beta_1(1)$)	0.284 (s.e. 0.070)	$[0.185,\ 0.388]$
Livingston cumulative loading, stable ($\beta_1(1)$)	0.302 (s.e. 0.060)	$[0.185,\ 0.410]$
MSC cumulative loading, volatile ($\beta_1(1) + \beta_2(1)$)	0.383 (s.e. 0.019)	$[0.329,\ 0.437]$
SPF cumulative loading, volatile ($\beta_1(1) + \beta_2(1)$)	0.242 (s.e. 0.040)	$[0.168,\ 0.317]$
Livingston cumulative loading, volatile ($\beta_1(1) + \beta_2(1)$)	0.193 (s.e. 0.044)	$[0.112,\ 0.271]$
MSC cumulative loading, regime shift ($\beta_2(1)$)	0.079 (s.e. 0.030)	$[0.033,\ 0.125]$

Notes: Estimates and prewhitened Newey–West standard errors are taken from Tables 6 and 9. Brackets: residual moving-block bootstrap (MBB) 95% percentile intervals from 4,999 replications.

package produce the eleven quantities of Table E.3 and agree digit for digit. Holding the regressors fixed, we resample residual blocks of one year (twelve monthly, four quarterly, or two semiannual observations) with replacement, rebuild the dependent variable from the fitted values plus the resampled residuals, and re-estimate the model on each of 4,999 samples. We report 95% percentile intervals. For each labeled joint test (L1–L8), the accompanying p_{MBB} is instead a bootstrap p-value from 1,999 replications. We draw residual blocks from the null-imposed model, the restricted least-squares fit under that test's constraints, and recompute the prewhitened Newey–West F statistic in each replication. Block start points are drawn uniformly with replacement, the concatenated series is truncated to the sample length, and bootstrap p-values use the convention $(1 + \#\{F_b \geq F_0\})/(1 + B)$, with the rare replication whose restricted re-estimation fails discarded.

Table E.3 reports the results. The bootstrap confirms both halves of the professional efficiency contrast and the formation results. The volatile-regime inefficiency intervals, the six cumulative-loading intervals, and the household loading-shift interval all exclude zero, while the stable-regime efficiency intervals straddle it. We also use the moving-block bootstrap, with the same one-year block lengths, for the pairwise loss-difference intervals of Table 3, with 9,999 replications, resampling the loss difference and the regime indicator jointly in blocks, which preserves the dependence between relative performance and the regime.

E.4 Parametric bootstrap for regime-estimation uncertainty

We estimate the regime indicator Regime_t in a first-stage maximum-likelihood fit of the Markov-switching model (2), so second-stage standard errors understate total uncertainty. The following parametric bootstrap quantifies the first-stage contribution.

We apply it to a small set of core quantities rather than to every coefficient:

1. Simulate a pseudo quarterly inflation path of the classification window's length from the estimated MS-AR(1), using the point estimates of $(\alpha_s, \rho_s, \sigma_s^2)$ and the estimated transition matrix, after a 100-quarter burn-in.

2. Re-estimate the model on the simulated path from two starting values, the values used for the actual-data estimation and the actual-data point estimates. Retain the fit with the higher likelihood and label its higher-variance state volatile.

3. Using the re-estimated parameters, recompute smoothed probabilities on the actual inflation series and reclassify the sample at the baseline cutoff of 0.5, rebuilding the monthly indicator by the same quarter-end inheritance rule as the baseline.

4. Re-run the second stage on the actual data with the reclassified regime indicator, and store the core quantities.

The core quantities are the four squared-loss differences by regime, the professional own-forecast coefficients in the stable regime and their volatile-regime sums, and the stable and volatile cumulative loadings of the three surveys with their regime shifts. Because we hold the data fixed and only the classification varies, the resulting intervals isolate first-stage uncertainty and complement, rather than replace, the sampling-based inference of the main text.

Table 13 in Section 6.4 reports the results from 999 replications. Estimation succeeded in all 999 replications, so we never used the rule that discards a replication whose estimation fails at both starting values or whose smoothing of the actual series fails. A quantity can still be undefined within a successful replication. When the reclassified indicator takes a single value over a survey's estimation sample, or leaves one regime too few dates to identify the regime interactions, the interaction coefficients drop out; we then treat the survey's regime-specific coefficients as undefined rather than keep a stable-regime coefficient that the volatile dates helped determine. The loss difference of a regime that never occurs likewise has no dates to draw on. That leaves each entry of Table 13 defined in between 978 and 999 replications. The volatile-regime loss differences are defined in all 999, and the smallest count, 978, belongs to the three Livingston loading rows.